

When Uncertainty Blows in the Orchard: Comovement and Equilibrium Volatility Risk Premia

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ABSTRACT

We provide novel evidence for an equilibrium link between investors' disagreement, the market price of volatility and correlation, and the differential pricing of index and individual equity options. We show that belief disagreement is positively related to (i) the wedge between index and individual volatility risk premia, (ii) the different slope of the smile of index and individual options, and (iii) the correlation risk premium. Priced disagreement risk also explains returns of option volatility and correlation trading strategies in a way that is robust to the inclusion of other risk factors and different market conditions.

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The pricing of index and individual stock options differs in a number of striking empirical dimensions, which represent a challenge to modern asset pricing models. First, as noted in Bakshi, Kapadia, and Madan (2003), the index option implied volatility smile is on average steeper and more negatively skewed than the smile of single stock options. Second, the average volatility risk premium of index options is typically larger than that of individual stock options. For instance, while the average at-the-money implied volatility of S&P 100 index options has been about 19.2% on a yearly basis, realized volatility has been only about 16.7%, which implies an index volatility risk premium of 2.5%. In contrast, the average volatility risk premium of single-name S&P 100 stocks has been only 0.9% in the same sample period. Third, such significant cross-sectional differences in compensation for volatility risk result in large risk adjusted returns for a number of option strategies. In our data, we find that dispersion portfolios that are short index volatility and long individual stock volatility can achieve an annualized Sharpe ratio of about 1.5 without an apparent exposure to standard risk factors.¹

Even though the stylized facts on volatility and correlation risk premia are well established, little is known about the structural economic mechanism behind them. In this paper, we motivate theoretically and test empirically a novel explanation for these links, based on the optimal risk-sharing of investors with heterogeneous beliefs. We specify a two-tree Lucas (1978) economy, in which belief disagreement is a priced equilibrium risk factor, and characterize the relation between the difference in opinions with (i) the different option skews of index and individual stocks, (ii) the volatility and correlation risk premia of index and individual options, and (iii) the risk-return profile of option volatility and correlation strategies. In our model, volatility and correlation are priced risk factors with state-dependent risk premia that are driven by heterogeneity in beliefs and the perceived level of investors' uncertainty.

Using the model solution, we derive a number of testable predictions. First, greater disagreement about future dividends induces a larger negative difference in the skews of index and individual option smiles, which reflects the higher price of insuring bad aggregate dividend states in our economy. Second, the greater the difference in beliefs, the larger the volatility risk premium of index options relative to the volatility risk premium of individual options. The difference between index and individual volatility risk premia is linked to a large correlation risk premium, which is positively related to the degree of aggregate information used by investors to form their dividend beliefs. Third, option volatility and correlation strategies based on straddle and dispersion portfolios generate attractive excess returns and Sharpe ratios,

with no apparent exposure to standard risk factors but with a significant beta with respect to systematic disagreement shocks.

Using monthly data on S&P 100 index and single-stock options for the period from January 1996 to June 2007, we test the predictions of our model. To compute proxies for belief disagreement for each individual firm in the S&P 100, we use analysts' earning forecasts from Institutional Brokers Estimate System (I/B/E/S). We then compute the cross-sectional weighted average of the individual disagreement proxies as a bottom-up proxy for a common component in the diversity of beliefs across firms. Our empirical analysis produces a number of novel results that support the model predictions.

First, we find that a single common disagreement component has strong predictive power for future firm-specific disagreement across stocks. This evidence is consistent with our modeling approach, in which sentiment on future firm growth prospects is generated by both firm-specific dividend news and aggregate dividend information. Second, individual and common disagreement proxies increase the difference between index and individual option skews in the data: A one-standard-deviation change in individual disagreement implies about a half-standard-deviation change in the difference between index and individual option slopes of the smile. Third, difference in beliefs significantly increases the wedge in the volatility risk premia of index and individual stock options, as well as the correlation risk premium, in a way that is robust to the inclusion of several other control variables: A one-standard-deviation shock in the individual disagreement implies almost a one-standard-deviation change in the difference between index and individual volatility risk premia, while common disagreement has the strongest explanatory power for correlation risk premia, even after controlling for other risk factors. Fourth, straddle and dispersion option strategies have returns with no significant economic exposure to standard risk factors, but a significant exposure to systematic disagreement shocks. There is a strongly significant difference of 15.9% (6.2%) between the monthly average returns of the quintile straddle (dispersion) portfolio with the lowest disagreement options and the quintile portfolio with the highest disagreement options. Fifth, using a two-pass Fama and MacBeth (1973) approach, we find that common disagreement carries a statistically significant price of risk that accounts for two-third (one-third) of the total cross-sectional variation in straddle (dispersion) portfolio returns.

Our paper borrows from different strands of the literature. The first studies equilibria with multiple assets. Menzly, Santos, and Veronesi (2004) and Santos and Veronesi (2010) solve an

economy with external habit and mean-reverting dividend shares. Pavlova and Rigobon (2007) investigate a two-country, two-good economy with demand shocks and log-linear preferences. In a two-tree Lucas (1978) economy, Cochrane, Longstaff, and Santa-Clara (2008) highlight the implications for stock returns and stock correlations. Martin (2013) introduces collections of Lucas trees (Lucas orchards) with rare disasters. Our focus is very different from that in this literature, because we explain stock return comovement and the market price of volatility and correlation risk by the risk-sharing behavior of disagreeing investors, rather than through direct market-clearing effects in an homogeneous economy. We test empirically the model predictions and show that priced disagreement risk does indeed play an essential role in understanding the equilibrium price of volatility and correlation risk.

Our work is closely related to the literature on volatility risk premia. Fleming, Ostdiek, and Whaley (1995), Jackwerth and Rubinstein (1996), and Christensen and Prabhala (1998), among others, observe that on average index volatilities are substantially lower than option implied volatilities, while Buraschi and Jackwerth (2001), Coval and Shumway (2001), and Bakshi and Kapadia (2003a) show that volatility risk is priced at the index level. Empirical evidence that volatility risk premia for single-name options are smaller than for index options is found in Bakshi and Kapadia (2003b), Duarte and Jones (2007), and Carr and Wu (2009), among others. Driessen, Maenhout, and Vilkov (2009) explain the difference between index and single-name variance risk premia by a large correlation risk premium. Goyal and Saretto (2009) find that long-short portfolios of straddles sorted with respect to the individual differences between implied and realized volatility can produce attractive returns, with no apparent exposure to standard risk factors. Since this evidence is difficult to explain within rational expectations models, Goyal and Saretto (2009) interpret their findings as an indication of option misspricing. Our paper reconciles these results in a structural equilibrium model with priced disagreement risk, which offers a single rational explanation for (i) the differential pricing of index and individual stock options, (ii) the relation between volatility and correlation risk premia, and (iii) the excess returns of option volatility and correlation strategies. Our empirical analysis supports the model predictions and our Fama and MacBeth results highlight the key role of priced disagreement risk for explaining the apparently contradicting evidence in the literature.

We are not the first to address the link between belief disagreement and option implied volatilities. Buraschi and Jiltsov (2006) show in a single-tree Lucas economy that large belief

disagreement can increase the implied volatility smile of index options. This paper differs from their work along several dimensions. We consider an economy with several stocks, which allows us to study in a single framework the structural link between belief heterogeneity and (i) the difference between index and individual equity volatility risk premia, (ii) the difference of index and individual implied volatility slopes, (iii) the correlation risk premium embedded in index options, and (iv) the risk-return tradeoff of option volatility and correlation portfolios. Empirically, we employ a structural proxy for the common disagreement, computed from the cross-section of individual differences in earning forecasts, and show that the model predictions are broadly consistent with the empirical evidence.

The rest of the paper is organized as follows. Section I provides the model setup. Section II derives the main theoretical model predictions. Section III describes our panel data set and Section IV presents the results of our empirical study. Finally, Section V concludes. For brevity, we defer all proofs to an Internet Appendix. The Internet Appendix is available in the online version of this article.

I. The Economy with Heterogeneous Beliefs

We borrow from the single-firm economies in Buraschi and Jiltsov (2006), Dumas, Kurshev, and Uppal (2009), and Buraschi, Trojani, and Vedolin (2013) and extend these models to a two-tree Lucas economy with heterogeneous beliefs, in which belief heterogeneity is related to dividend shocks and another source of independent news about aggregate economic growth. In this setting, we obtain a number of testable empirical predictions for the differential pricing of index and individual options and for the wedge between index and individual option volatility risk premia.

A. State Dynamics

Uncertainty is driven by a Brownian motion $W = \left(W_{D_i}, W_z, W_{\mu_{D_i}}, W_{\mu_z}\right)'_{i=1,2}$ on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$. Two firms, indexed by $i = 1, 2$, produce their perishable good and pay a dividend process with dynamics $d \log D_i(t) = \mu_{D_i}(t)dt + \sigma_{D_i}dW_{D_i}(t)$, where $\sigma_{D_i} > 0$. The expected growth rates of dividends $\mu_{D_i}(t)$ are unobservable and follow the dynamics

$$d\mu_{D_i}(t) = (a_{0D_i} + a_{1D_i}\mu_{D_i}(t))dt + \sigma_{\mu_{D_i}}dW_{\mu_{D_i}}(t) .$$

Parameter $\sigma_{\mu_{D_i}} > 0$ measures the economic uncertainty generated by the stochastic properties of firm i 's individual growth rate.

In reality, investors are exposed to many potential sources of news, which can be relevant to form their beliefs about the future prospects of a firm. Some of this information can be related to firm characteristics, while other news can have a more systematic or aggregate character, producing information about the growth rate of the economy as a whole. Since the degree of common information used by investors to form their beliefs has implications for the comovement of beliefs about firms' future prospects, we embed this feature into our model by a signal $dz(t)$ related to the dividend process of both firms:

$$\begin{aligned} dz(t) &= (\alpha_{D_1}\mu_{D_1}(t) + \alpha_{D_2}\mu_{D_2}(t) + \beta\mu_z(t)) dt + \sigma_z dW_z(t), \\ d\mu_z(t) &= (a_{0z} + a_{1z}\mu_z(t)) dt + \sigma_{\mu_z} dW_{\mu_z}(t), \end{aligned}$$

where $\sigma_z > 0$. The variable $dz(t)$ can be interpreted as an independent noisy signal of aggregate economic growth. If $\alpha_{D_1} = \alpha_{D_2} = 0$, dividends and signals are independent and investors exclusively use dividend information to form their beliefs. If $\alpha_{D_1} \neq 0$ or $\alpha_{D_2} \neq 0$, then $E_t(dz(t))$ is correlated with expected dividend growth and $dz(t)$ generates useful independent information for estimating dividend growth. Since $E_t(dz(t)) \neq \alpha_{D_1}\mu_{D_1}(t) + \alpha_{D_2}\mu_{D_2}(t)$ when $\beta \neq 0$, the information generated by $dz(t)$ is not redundant and uncertainty about the difference $E_t(dz(t)) - (\alpha_{D_1}\mu_{D_1}(t) + \alpha_{D_2}\mu_{D_2}(t))$ impacts the belief formation process. Notice that heterogeneity in the uncertainty parameter σ_{μ_z} between agents implies that investors attribute different weight to signal news in their belief formation process.

When $\alpha_{D_1} = \alpha_{D_2}$, $dz(t)$ has the interpretation of a signal of aggregate dividend growth, because it produces a noisy estimate of $\mu_{D_1}(t) + \mu_{D_2}(t)$. As such, it also helps predict future expected dividend growth. Different choices of parameters β and α_{D_i} determine the relation between $E_t(dz(t))$ and $\mu_{D_1}(t) + \mu_{D_2}(t)$, which is stronger as α_{D_i}/β increases. This relation can also be interpreted as a parsimonious parametrization of rational inattention features, in which a lower ratio β/α_{D_i} parameterizes higher inattention. Rational inattention has been studied by a large literature in psychology and finance, and has been shown to be linked to agents' reaction to aggregate economic news; see Peng and Xiong (2006) and Peng, Xiong, and Bollerslev (2007), among others.

B. Specification of Disagreement

Two investors, indexed by $n = A, B$, update their beliefs following Bayes' rule and a standard Kalman-Bucy filter. To write the model dynamics in vector form, denote investor specific parameters by $b^n = \text{diag}(\sigma_{\mu_{D_1}}, \sigma_{\mu_{D_2}}, \sigma_{\mu_z}^n)$ and parameters that are common across investors by $a_0 = (a_{0D_1}, a_{0D_2}, a_{0z})'$, $a_1 = \text{diag}(a_{1D_1}, a_{1D_2}, a_{1z})$, $B = \text{diag}(\sigma_{D_1}, \sigma_{D_2}, \sigma_z)$, and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \alpha_{D_1} & \alpha_{D_2} & \beta \end{pmatrix}. \quad (1)$$

The heterogeneity in uncertainty parameter $\sigma_{\mu_z}^n$ models the investor-specific perception of the noisiness of signal $dz(t)$, which affects investors' Bayesian belief updating and thus their future dividend disagreement. Denote by $Y(t) := (\log D_1(t), \log D_2(t), z(t))'$ the observable state vector, by $\mathcal{F}_t^Y$ the information generated up to time t , and by $E^n(\cdot)$ the expectation operator under the subjective probability of investor n . The posterior belief for state $\mu(t) := (\mu_{D_1}(t), \mu_{D_2}(t), \mu_z(t))'$ is $m^n(t) := E^n(\mu(t)|\mathcal{F}_t^Y)$. Its posterior covariance matrix solves a matrix Riccati differential equation, with a solution that can depend on the initial prior. We consider the steady-state solution of this equation, which is independent of agents' prior.

The posterior belief dynamics of agent A follows as: $dm^A(t) = (a_0 + a_1 m^A(t))dt + \gamma^A A' dW_Y^A(t)$, with initial conditions $m^A(0) = m_0^A$, where $dW_Y^A = (dW_{D_1}^A, dW_{D_2}^A, dW_z^A)'$ is the standardized innovation process from the perspective of investor A and γ^A the corresponding posterior covariance matrix.² Note that if matrix (1) is diagonal ($\alpha_{D_1} = \alpha_{D_2} = 0$), then the three-dimensional vector $m^A(t)$ has independent components and dividend beliefs are independent of the signal dynamics. However, if A is not diagonal and $\beta \neq 0$, signal $dz(t)$ produces non-redundant information about dividend growth. This information makes dividend beliefs dependent on signal shocks, so that signal belief $E^n(dz(t)|\mathcal{F}_t^Y)$ predicts future dividend beliefs $E^n(\mu_{D_1}(s)|\mathcal{F}_s^Y)$ and $E^n(\mu_{D_2}(s)|\mathcal{F}_s^Y)$, $s \geq t$.

Disagreement about dividends and the marketwide signal can be summarized by the vector $\Psi(t) := (\Psi_{D_1}(t), \Psi_{D_2}(t), \Psi_z(t))' := B^{-1}(m^A(t) - m^B(t))$. The first two components of $\Psi(t)$ capture disagreement about dividend growth rates, while the third is proportional to disagree-

ment about the noisy component $E_t(dz(t)) - (\alpha_{D_1}\mu_{D_1}(t) + \alpha_{D_2}\mu_{D_2}(t)) = \beta\mu_z(t)$ in $E_t(dz(t))$. Using Itô's Lemma, the dynamics of $\Psi(t)$ are given by

$$d\Psi(t) = B^{-1} (a_1 B + \gamma^B A' B^{-1}) \Psi(t) dt + B^{-1} (\gamma^A - \gamma^B) A' dW_Y^A(t), \quad (2)$$

where γ^B is investor B 's posterior covariance matrix. In equation (2), $dW_Y^A(t)$ loads on $d\Psi(t)$ proportionally to

$$(\gamma^A - \gamma^B) A' dW_Y^A(t) = (\gamma^A - \gamma^B) \begin{pmatrix} dW_{D_1}^A(t) + \alpha_{D_1} dW_z^A(t) \\ dW_{D_2}^A(t) + \alpha_{D_2} dW_z^A(t) \\ dW_z^A(t) \end{pmatrix}.$$

When $\alpha_{D_1} = \alpha_{D_2} = 0$, $\gamma^A - \gamma^B$ is a diagonal matrix and dividend disagreement is independent of signal shocks. More generally, when $\alpha_{D_1} \neq 0$ and $\alpha_{D_2} \neq 0$, $\gamma^A - \gamma^B$ is a full matrix and signal shocks generate a common component in the dividend disagreement across firms, which is larger when $\alpha_{D_1} = \alpha_{D_2}$. In this way, our model motivates a latent common factor in the multivariate dividend disagreement process, generated by the aggregate signal used by investors to form dividend beliefs. It follows that even if dividends are weakly dependent, a comovement in dividend disagreement processes can arise whenever $dz(t)$ is perceived to produce non-redundant information about aggregate dividend growth. Importantly, this implies that current disagreement about the noisy component of aggregate growth, $\Psi_z(t)$, is a predictive variable for future dividend disagreements $\Psi_{D_1}(s)$ and $\Psi_{D_2}(s)$, $s \geq t$.

In the data, proxies for earnings disagreement often display a large countercyclical common component that exhibits distinct spikes around crisis periods. To study this link in the context of our model in more detail, it is useful to look at the properties of the conditional distribution of $\Psi_{D_1}(t)$ and $\Psi_{D_2}(t)$ in our model. In the comparative statics of Figure 1, we find that higher average subjective uncertainty on the aggregate signal (parameterized by $\bar{\sigma}_{\mu_z} \equiv \frac{1}{2} (\sigma_{\mu_z}^A + \sigma_{\mu_z}^B)$) mostly increases average dividend disagreement, while higher uncertainty heterogeneity (parameterized by $\Delta\sigma_{\mu_z} \equiv \sigma_{\mu_z}^B - \sigma_{\mu_z}^A$) tends to increase dividend disagreement volatility. This result follows from the dependence of the (Gaussian) distribution of $\Psi(t)$ on the investor-

specific uncertainty parameter $\sigma_{\mu_z}^n$, $n = A, B$, which is explicit in our model through the link between posterior variances γ^n and $\sigma_{\mu_z}^n$

$$\Psi(t)|\Psi(0) \sim \mathcal{N}\left(e^{tM}\Psi(0), te^{tM}B^{-1}(\gamma^A - \gamma^B)A'BB'^{-1}A(\gamma^A - \gamma^B)B^{-1}e^{tM}\right),$$

where $M = B^{-1}(a_1B + \gamma^BA'^{-1})$.

Figure 1 about here

To compute comparative statics, we assume for simplicity a symmetric economy calibrated to the S&P 500 dividend dynamics. In the left panel of Figure 1, average dividend disagreement smoothly increases from 0.054 to 0.056 as the parameter $\bar{\sigma}_{\mu_z}$ increases from 0 to 0.1. For comparison, in the data the average disagreement across all firms is about 0.049, while it is about 0.021 (0.095) for low (high) book-to-market firms. In the right panel of Figure 1, the volatility of the dividend disagreement rises from zero to about 0.35 as $\Delta\sigma_{\mu_z}$ increases from zero to 0.01. In our sample, the disagreement standard deviation of all firms is about 0.017, while it is about 0.0009 (0.063) for low (high) book-to-market firms. Finally, we also find that the first moment of the dividend disagreement increases with the current level of $\Psi_z(0)$, that is, $\Psi_z(0)$ is a predictive variable for future dividend disagreement.

C. Comovement of Belief Disagreement

We can now study the comovement features of $\Psi_{D_1}(t)$ and $\Psi_{D_2}(t)$ in our model in more detail using the analytical correlation expressions provided in the Internet Appendix. Figure 2 presents the comparative statics of the model-implied correlation between $d\Psi_{D_1}$ and $d\Psi_{D_2}$ with respect to parameters α_{D_1} and α_{D_2} .³

Figure 2 about here

According to intuition, $\rho(d\Psi_{D_1}(t), d\Psi_{D_2}(t))$ is an increasing function of α_{D_1} and α_{D_2} , and attains a maximum for $\alpha_{D_1} = \alpha_{D_2}$, that is, when signal shocks $dz(t)$ imply large belief revisions about aggregate dividend growth. At the calibrated parameters, this excess comovement can be large: for $\alpha_{D_1} = \alpha_{D_2} = 0.4$ it can be as high as 30%, depending on $\sigma_{\mu_z}^A$ and $\sigma_{\mu_z}^B$. The correlation $\rho(d\Psi_{D_1}(t), d\Psi_{D_2}(t))$ is also linked to the structure of the individual uncertainty parameters

$\sigma_{\mu_z}^n$ across agents: as either the average uncertainty $\bar{\sigma}_{\mu_z}$ or the uncertainty heterogeneity $\Delta\sigma_{\mu_z} \equiv \sigma_{\mu_z}^A - \sigma_{\mu_z}^B$ increase, the correlation increases because signal shocks imply larger revisions of dividend beliefs for all agents.

Given the empirical evidence that investors tend to focus on aggregate information during crisis periods, our model suggests a structural channel for explaining the countercyclical comovement of disagreement proxies across firms. If the average perceived uncertainty or the uncertainty heterogeneity across investors also tend to increase during crisis times, these countercyclical comovement features can be even stronger.

D. Equilibrium

We derive testable implications for the relation between belief disagreement and index and individual options volatility risk premia using a simple two-tree exchange economy with two consumption goods. Subject to the budget constraint, investors maximize their life-time expected utility,

$$V^n = \sup_{c_{D_1}^n, c_{D_2}^n} E^n \left(\int_0^\infty e^{-\delta t} \left(\frac{c_{D_1}^n(t)^{1-\gamma}}{1-\gamma} + \frac{c_{D_2}^n(t)^{1-\gamma}}{1-\gamma} \right) dt \mid \mathcal{F}_0^Y \right), \quad (3)$$

where $c_{D_i}^n(t)$ is the consumption of agent n of good i and $\gamma, \delta > 0$ are the common relative risk aversion and time preference parameters.

By allowing each tree to produce its own fruit, we introduce an imperfect substitution rate between goods. Representative-agent models studying multiple trees with a single fruit include, among others, Menzly, Santos, and Veronesi (2004), Cochrane, Longstaff, and Santa-Clara (2008), Chen and Joslin (2012), and Martin (2013). In these economies, comovement of returns can arise through market-clearing effects. Such effects are weak in our setting because of the imperfect substitution between goods. In our model, return comovement arises due to the equilibrium risk-sharing policies of investors with diverse beliefs.

To identify a unique equilibrium stochastic discount factor, we assume that agents can trade in a risk-free bond, in zero net supply, shares of the two firms, in positive supply, and equity and index options, all in zero net supply. The equilibrium is solved using standard

martingale methods and produces closed-form expressions for the stochastic discount factor of agent n , denoted by $\xi^n(t)$, which is a function of a stochastic weighting process $\lambda(t)$:⁴

$$\xi^A(t) = \frac{e^{-\delta t}}{y_A} D_1(t)^{-\gamma} \left(1 + \lambda(t)^{1/\gamma}\right)^\gamma, \quad \xi^B(t) = \frac{e^{-\delta t}}{y_B} D_1(t)^{-\gamma} \left(1 + \lambda(t)^{1/\gamma}\right)^\gamma \lambda(t)^{-1}, \quad (4)$$

where y_A and y_B are Lagrange multipliers in the (static) budget constraint of agent A and B , respectively, and $\lambda(t) := y_A \xi^A(t) / (y_B \xi^B(t))$ follows the dynamics

$$\frac{d\lambda(t)}{\lambda(t)} = - \left(\sum_{i=1}^2 \Psi_{D_i}(t) dW_{D_i}^A(t) + \left(\sum_{i=1}^2 \alpha_{D_i} \Psi_{D_i}(t) \frac{\sigma_{D_i}}{\sigma_z} + \beta \Psi_z(t) \right) dW_z^A(t) \right). \quad (5)$$

The dynamics of $\lambda(t)$ depends on perceived shocks $W_{D_1}^A(t)$, $W_{D_2}^A(t)$, and $W_z^A(t)$, which load on $\lambda(t)$ proportionally to $\Psi_{D_1}(t)$, $\Psi_{D_2}(t)$, and $\sum_{i=1}^2 \alpha_{D_i} \Psi_{D_i}(t) \frac{\sigma_{D_i}}{\sigma_z} + \beta \Psi_z(t)$, the degree of disagreement about $E_t(dz(t))$. The dependence of state prices on signal shocks $dW_z^A(t)$ increases when the relative precision of signals and dividends increases (σ_{D_i}/σ_z is large) or when the signal is more strongly related to expected dividend growth (α_{D_1} and α_{D_2} increase).

State prices feature an endogenous stochastic volatility, which is increasing with respect to all disagreement proxies. Thus, option prices and volatility risk premia also directly depend on the belief disagreement. The intuition is simple. State prices reflect the desire of pessimistic investors to ensure a satisfactory consumption level in bad states. Since investors' expected marginal utility must equate in each state, the consumption share of the pessimistic investor is countercyclical and the protection seller (the optimist) must be compensated ex-ante by a premium, which is state dependent because it is a function of the degree of heterogeneity in beliefs.

Financial protection can be bought through individual stock or index put options. The first contract gives protection against a low dividend of a single firm, the second contract against a low aggregate dividend. If dividends are perceived to be more correlated, low aggregate consumption states are more likely and protection against bad aggregate dividend states becomes relatively more expensive. Thus, the protection premium embedded in out-of-the-money put index options gets larger than the one implied by individual put options. Through this mechanism, the model can create a wedge between the pricing of index and individual stock options, which can help to explain the observed difference in volatility risk premia of index and individual option markets.

E. Pricing

Given the equilibrium state price densities, we can price any contingent claim in our economy.⁵ State prices are functions of $D_1(t)$, $D_2(t)$, and $\lambda(t)$. Therefore, the joint density of these variables is needed for pricing purposes. This density is not analytically tractable, but its Laplace transform can be computed in closed form. Borrowing from the solution techniques in Dumas, Kurshev, and Uppal (2009), we can characterize this Laplace transform. The resulting pricing formulas are collected in Lemma 3 of the Internet Appendix. Equilibrium stock and index prices are in semi-closed form, up to a numerical integration. This feature largely reduces the computational costs of the equilibrium. Option prices, which are derivatives written on equilibrium stock and index values, require a Monte Carlo simulation step in their computation.

II. Model Predictions

Using the pricing formulas of Lemma 3 in the Internet Appendix, we can study the empirical predictions for (i) the market price of variance-covariance risk and (ii) the differential pricing of individual and index options, which are functions of the belief disagreement and the degree of aggregate signal uncertainty in our model. Motivated by the intuition developed earlier, we study comparative statics of equilibrium variables with respect to disagreement proxy $\Psi(t)$ and uncertainty parameters $\bar{\sigma}_{\mu_z}$ and $\Delta\sigma_{\mu_z}$. The parameters of the fundamentals $(\mu_{D^i}, \sigma_{D^i})$ are calibrated to match the sample mean and volatility of the dividends on the S&P 500 for the period 1996 to 2006. We assume that the volatility of the signal, σ_z , is as big as that of dividends. At the steady-state, we set the difference in economic uncertainty $(\sigma_{\mu_z}^A - \sigma_{\mu_z}^B)$ to -0.01. The average economic uncertainty $(\bar{\sigma}_{\mu_z})$ is fixed at 0.02. We choose the parameters such that the first two moments of the model-implied disagreement process approximately match the first and second moments of the proxy for common disagreement that we construct in the empirical section. All calibrated parameters are collected in Table I.

Table I about here

A. The Link Between Stock Return Correlation and Common Disagreement

In our economy, stock returns can be correlated even if dividends are only weakly linked. In contrast to representative-agent economies, such as Cochrane, Longstaff, and Santa-Clara (2008), market-clearing effects are not the key driver of endogenous stock comovement. Rather, it is the interdependence of aggregate informational effects coupled with optimal risk-sharing among agents that produces the main channel for comovement of (i) belief disagreement across stocks and (ii) stock returns. The first channel is potentially completely belief-driven: rational inattention behavior or higher economic uncertainty produce larger comovement in beliefs, which increases the aggregate cash-flow risk perceived by investors. The second channel works through endogenous risk-sharing across agents. In equilibrium, the more pessimistic agent consumes a higher fraction of aggregate consumption in bad consumption states. Therefore, she tries to reduce the exposure to stocks with a dividend process correlated with aggregate consumption. Those stocks are sold to the more optimistic investor at a discount. When aggregate signals increase agents' perception of the degree of dividend correlation, the price adjustment needed for ex-ante optimal risk-sharing across agents implies a higher discount for both stocks. This risk-sharing induced channel of comovement is absent in Lucas economies with homogeneous agents.

We can now solve for the conditional variances and covariance of stock returns. Figure 3 presents the comparative statics of the stock return correlation.⁶

Figure 3 about here

Consistent with intuition, the stock correlation is increasing with the degree of disagreement about dividends and the aggregate signal. In the absence of disagreement, the correlation is small, confirming the minor role of comovement dynamics driven by market-clearing effects in our economy. As expected, when aggregate uncertainty and uncertainty heterogeneity rise ($\bar{\sigma}_{\mu_z}$ or $\Delta\sigma_{\mu_z}$ increase), the stock correlation is higher. Return correlation also rises when rational inattention behavior is stronger (α_{D_1} and α_{D_2} increase).

B. The Differential Pricing of Index and Individual Option Smiles

A second key feature of the model is that it endogenously generates a steeper smile for index options than for individual stock options. Bakshi, Kapadia, and Madan (2003) link

the differential pricing of these options to their different risk-neutral skewness. In our model, risk-neutral skewness is generated by the asymmetric response of equilibrium state prices to dividend shocks. Since the more optimistic agent sells financial protection at a premium, the price of low dividend states increases with the degree of investors' disagreement. Additivity of investors' utility functions implies that asymmetric state price adjustments are amplified when pricing aggregate dividend states: the risk-neutral skewness of the index is more negative than for dividends of individual firms. As a consequence, index returns feature more negative skewness than stock returns. When dividends are perceived as more correlated under agents' beliefs, the higher likelihood of low aggregate dividend states further increases the difference between index and single-stock risk-neutral skewness, leading to the steeper smile for index options. The stronger negative skewness of index returns is directly reflected in Figure 4 by a steeper smile of index options. Note that as $\Psi_{D_1}(t)$ or $\Psi_z(t)$ increase, index option smiles become proportionally steeper than individual stock smiles, where this effect is larger in the presence of stronger rational inattention features or a higher market-wide uncertainty. We summarize these findings by the following prediction.

Figure 4 about here

Model Prediction I: Larger belief heterogeneity increases the (negative) difference between index and individual slopes of the option smile, where the difference is more pronounced in the presence of a larger common component in the cross-section of individual disagreement about firm cash-flows.

In our empirical analysis, we take Model Prediction I systematically to the data and test what fraction of the difference between index and individual option skews can be explained by common and individual disagreement proxies extracted from I/B/E/S data on professional earning forecasts.

C. Volatility and Correlation Risk Premia

The third key feature of the model is that correlation and volatility risk are priced in equilibrium. Is it well known that the difference between implied and realized volatility is naturally linked to the volatility risk premium of the option's underlying; see, for example, Carr and Wu (2009). Similarly, the larger volatility risk premium of index options can be

explained by a correlation risk premium embedded in the index constituents; see, among others, Driessen, Maenhout, and Vilkov (2009).⁷ Our model therefore motivates a potential tight link between commonality in dividend disagreement across firms, the larger volatility risk premium of index options, and the implied correlation risk premium of the index basket.

The correlation (variance) risk premium is the expected payoff of a correlation (variance) swap that pays the average realized correlation between stocks (realized variance) over a given investment horizon at a price, minus the correlation (variance) swap rate, paid at the maturity of the contract. Thus, the correlation (variance) risk premium is simply the difference in the expected average realized correlation between stocks (the expected realized variance of an asset) over a given investment horizon, under the physical and the risk-neutral probability. The volatility risk premium is simply the difference in the square root of the expected realized variance of an asset under the physical and the risk-neutral probability.

In our economy, a negative correlation risk premium arises because large correlation states are more likely under the risk-neutral distribution, as disagreement increases. It is the equilibrium risk-sharing that increases stock correlations when disagreement among investors rises, because low aggregate consumption states require a larger equilibrium reallocation of financial protection.

To compute observable proxies for the return of correlation and variance swaps in our model calibrations and our empirical analysis, we follow standard procedures. First, we compute the correlation (variance) swap payoff simply as the average realized daily correlation in the given basket of stocks (the realized variance of an asset) over a one-month holding period. Second, we infer correlation and variance swap rates from the cross-section of index and individual stock option implied volatilities with one-month maturity. More precisely, we follow Carr and Madan (1998), Britten-Jones and Neuberger (2000), and Carr and Wu (2009), among others, to infer the index and individual stock variance swap rates from the cross-section of option prices. From these, we can compute the correlation swap rate, which is a simple function of the index and individual variance swap rates. Figure 5 summarizes the main predictions of our model for the relation between volatility risk premia, correlation risk premia and belief heterogeneity.

Figure 5 about here

Individual volatility risk premia and the correlation risk premium are larger, in absolute value, as disagreement and economy-wide uncertainty are more pronounced; see the left and right panels of Figure 5, respectively. For instance, the correlation risk premium increases from nearly zero to about 1% as the dividend and the aggregate signal disagreement rise: the largest increases are caused by shocks in the aggregate disagreement component, which tend to be linked more strongly to stock correlation shocks. In summary, we obtain the following model prediction.

Model Prediction II: Higher heterogeneity in beliefs implies a larger difference between index and individual volatility risk premia (a larger correlation risk premium).

We test empirically Model Prediction II by estimating the fraction of (i) correlation risk premium and (ii) difference between index and individual variance risk premia that can be explained by empirical proxies for common and individual disagreement.

D. The Excess Returns of Option Volatility and Correlation Strategies

Our model features a nonzero risk premium for disagreement risk that implies novel predictions for the excess returns of options trading strategies. In particular, we can investigate the extent to which the abnormal return patterns documented in previous empirical literature can be explained by a risk premium for disagreement risk.

One of the most studied option volatility strategies is based on portfolios of straddles, which are combined long and short positions in a call and a put with identical moneyness and underlying. Straddles imply positive (negative) expected excess returns when the variance risk premium is positive (negative) and have been extensively analyzed empirically; see, among others, Coval and Shumway (2001) and Driessen and Maenhout (2007).

A well-known option correlation strategy is based on dispersion portfolios, that is, combined positions in a short straddle of index options and a long straddle portfolio of individual stock options. The short index straddle earns the index volatility risk premium. The long portfolio of individual straddles pays an average of individual volatility risk premia. Since the difference between index and individual volatility risk premia is negative on average, dispersion portfolios imply a positive expected excess return because of the negative correlation risk premium.⁸

Previous literature often highlights that it is difficult to reconcile the excess returns of volatility or correlation strategies with traditional risk factors; see, for example, Goyal and

Saretto (2009). Our model suggests a simple alternative explanation based on the role of priced disagreement risk: cheap options are those with a low degree of disagreement. When disagreement is large, the absolute difference between implied and realized volatility of the firm tends to be large as well, because investors need to be compensated for holding disagreement risk. It follows that if we sort the long side of a dispersion portfolio with respect to the dividend disagreement of the individual option straddles, we should obtain sorted expected excess returns. In the same way, the spread between sorted dispersion portfolios with respect to high and low disagreement should yield a positive expected excess return. Finally, since these returns reflect a premium for exposure to disagreement risk, they should be explained by risk factors related to (common) disagreement components. We verify these predictions in the context of our calibrated model.

To be broadly consistent with the main features of our data, we simulate return time series of 11 years and five months (2,877 days) by computing option returns using Lemma 3 of the Internet Appendix. At the beginning of each month, at-the-money index straddles are sold and at-the-money straddles on one of the two individual stocks are bought, in a way that makes the portfolio vega neutral with respect to the individual stock volatility. To construct individual straddles and the sorted dispersion portfolios, we consider both the stock with the lowest and the stock with the highest dividend disagreement at the beginning of each month. The position is held for one month and is dynamically delta hedged by buying a corresponding amount of the individual stock and investing the remainder in the risk-free bond. We obtain simulated time series of 137 monthly returns of sorted dispersion trades. We also study the returns of a short index put strategy, which is known to generate high returns and Sharpe ratios in the data. This strategy is a natural benchmark for our sorted dispersion portfolios. Table II summarizes our findings.

Table II about here

Dispersion portfolios sorted with respect to the lowest disagreement yield the most attractive risk-return (mean-variance) profile. Their annualized Sharpe ratio is approximately 1.9; all dispersion portfolios provide a larger Sharpe ratio than both short index puts and sorted straddle portfolios of individual stocks. Compared to short index puts, they also feature a lower kurtosis and negative skewness, indicating slightly lower tail risk. In contrast to

dispersion strategies, sorted straddle returns feature no pronounced tail risk exposure and, in particular, a positive skewness.

In our simulations, we also test whether these excess returns are indeed explained by common disagreement risk components, where we proxy for common disagreement risk by the weighted average of the dividend disagreements $\Psi_{D_1}(t)$ and $\Psi_{D_2}(t)$. To this end, we regress option excess returns on (i) the market return, (ii) the monthly change in the common disagreement proxy, and (iii) a nonlinear common disagreement effect modeled by the squared change in the common disagreement proxy. Estimated coefficients of these factor regressions are reported in the last four rows of Table II.

We find that all volatility and correlation option strategy returns have zero market beta, while the short index put has positive market exposure, as expected; see, for example, Broadie, Chernov, and Johannes (2007). All option strategies have positive beta with respect to the common disagreement proxy. Strategies with the highest disagreement beta are, respectively, the dispersion portfolios sorted with respect to low and high disagreement, the short index put, and the sorted portfolio of individual straddles. Finally, we observe weak evidence of a negative link with squared common disagreement shocks, highlighting potential nonlinear dependence of option excess returns on disagreement risk. In summary, we obtain the following model prediction.

Model Prediction III: The returns of sorted straddle and dispersion portfolios feature zero beta with respect to market risk, but positive beta with respect to common disagreement.

Empirically, we test Model Prediction III by analyzing whether the excess returns of straddle and dispersion portfolios feature systematic exposure to empirical proxies for common disagreement, in the time series or in the cross-section, but no exposure to standard risk factors like the market return.

E. Asymptotic Survival

An important argument supporting the plausibility of the rational expectations theory is the market selection hypothesis: agents holding systematically incorrect beliefs will eventually be driven out of the market. In the long run, only agents holding more accurate beliefs will survive so that equilibrium asset prices will be determined only by those agents with rational beliefs. Under the assumption of no exit and/or entry of new agents in the economy, models

with homogeneous preferences and heterogeneous beliefs are exposed to the market selection hypothesis criticism. Several results in the literature show, however, that the time required for asymptotic convergence in economies similar to ours is very long (see Dumas, Kurshev, and Uppal (2009)).

Moreover, the market selection hypothesis is not robust to simple extensions. Yan (2008) studies this problem in the context of CRRA preferences and characterizes the conditions for survival of agents holding incorrect beliefs. He shows that survival depends on the accuracy of the beliefs, the patience parameter, and the risk aversion coefficient of the agent. Indeed, in absence of heterogeneity in preferences, those agents with incorrect beliefs cannot survive in the limit. The speed of the selection process depends on the assumption about agents' learning and it is very slow in some calibrated examples.

If preferences vary across investors even slightly, however, then it becomes possible for an irrational investor to dominate the market, even if his beliefs persistently and substantially deviate from the truth. Borovička (2012) extends this analysis to the case of homothetic recursive preferences of the Duffie-Epstein-Zin type. He shows that when risk aversion is larger than the inverse of the intertemporal elasticity of substitution, equilibria can arise in which both agents survive with strictly positive probability.

III. Data

To test empirically the main predictions of our model, we build a monthly merged panel data set, consisting of option and stock prices on all constituents of the S&P 100 index, the time series of index spot and option prices, and analysts' forecasts of future earnings from the I/B/E/S database. The sample period of our study is from January 1996 to June 2007.

Options Data: We use option information from the OptionMetrics Ivy DB database. Our option sample contains trades and quotes of S&P 100 index options and all individual options on all constituents of the S&P 100, traded on the Chicago Board Options Exchange (CBOE). The S&P 100 is a capitalization-weighted index with quarterly rebalancing. Options on the index are European style and expire on the third Friday of the contract month. Individual stock options are American style and usually expire on the Saturday following the third Friday of the contract month. The Internet Appendix contains the list of data filters we apply to the option data, as well as details on how we calculate implied volatilities and correlations.

Stock Return Data: In order to calculate the realized volatility of the index and individual stocks, we use daily returns from CRSP for single stocks and from OptionMetrics for the index. We calculate the realized volatility over 21-day windows, requiring that the stock has at least 15 nonzero return observations within each window.

Proxies for Differences in Beliefs: We use standard measures of belief disagreement using forecast data from the I/B/E/S database. For each firm i in each month t , we calculate the mean absolute standard deviation and consensus estimate of analyst forecasts of earnings per share from the unadjusted I/B/E/S database. Firm-specific disagreement is then defined as the ratio between the consensus estimate and the mean absolute standard deviation. Common disagreement is defined as the market capitalization weighted average of all firms. Market capitalization data are taken from the COMPUSTAT database. To circumvent the problem of using stock-split adjusted data, as described, for example, in Diether, Malloy, and Scherbina (2002), we use unadjusted data. We extend each forecast date to its revision date and if an analyst makes more than one forecast per month, we take the last forecast that was confirmed.⁹

Other Control Variables: It is well known that cross-sectional differences in option implied volatilities are related to firm-specific CAPM betas (see, for example, Chang, et al. (2011)). To control for such effects, we compute monthly conditional betas, using historical returns over a window of 180 days, and include them as explanatory variables in our regressions.

We control for liquidity effects using two different proxies. To control for option liquidity effects, we calculate the ratio of option trading volume and the number of shares in the underlying outstanding. To account for equity market liquidity effects, we use the Pástor and Stambaugh (2003) measure of liquidity.

In volatility risk premium regressions, we control for jump risk by explicitly proxying for differences in risk neutral skewness across stocks. We proxy risk neutral skewness with the difference between the implied volatility of a put with 0.92 strike-to-spot ratio (or the closest available) and the implied volatility of an at-the-money put, divided by the absolute difference in strike-to-spot ratios.

An alternative explanation for the emergence of volatility risk premia of single-stock and index options is the demand-based argument of Bollen and Whaley (2004). We therefore consider buying pressure as an additional explanatory variable in our regressions. Following Bollen and Whaley (2004), we define net buying pressure as the number of contracts traded

during the month at prices higher than the prevailing bid-ask quote midpoint, minus the number of contracts traded during the month at prices below the prevailing bid-ask quote midpoint, multiplied by the absolute value of the option’s delta and scaled by the total trading volume across all option series.

An additional alternative hypothesis for a time-varying endogenous correlation is suggested by Shiller (1984) and Shleifer and Summers (1990). They argue that the dynamic interaction of noise traders and rational arbitrageurs can lead to price formation processes in which arbitrage activity may not fully absorb correlated demand shocks. In this context, investor sentiment may induce endogenous (realized) return correlation. We control for this effect and quantify the different roles played by differences in beliefs versus investor sentiment. To this end, we follow Baker and Wurgler (2007) and proxy sentiment by the first principal component of trading volume (as measured by NYSE turnover), the dividend premium, the closed-end fund discount, the number and first-day returns on IPOs and the equity share in new issues; see Baker and Wurgler (2007).¹⁰

Finally, we control for other standard risk factors in the literature using the Fama and French (1993) and momentum factors. Business cycle effects are captured by a number of other variables, including the market price-to-earning ratio, industrial production, housing starts, the producer price index, and nonfarm employment.¹¹ Our macro proxy is the first principal component estimated from these variables.

IV. Empirical Analysis

In this section, we first study the link between firm-specific and common disagreement proxies in the data. We test empirically the predictions of our model. Using panel regression methods, we first address Model Predictions I and II by studying the extent to which individual and common belief disagreement proxies help to explain (i) differences between index and individual stock volatility risk premia, (ii) differences in the slope of the smile of index and individual stock options, and (iii) correlation risk premia. We then focus on Model Prediction III and analyze whether there is evidence that disagreement risk is priced in the time series and the cross-section of option returns. In the time series, we estimate the beta of the excess returns of straddle and dispersion strategies with respect to innovations in common disagreement proxies. In the cross-section, we test whether cross-sectional differences in risk-adjusted returns

are explained by their exposure to disagreement. Finally, we quantify the price of disagreement risk in option returns based on a Fama and MacBeth (1973) approach.

A. *Long-Run Returns and Higher-Order Disagreement*

In our model, disagreement related to the aggregate marketwide signal, $\Psi_z(t)$, is a predictive variable for future dividend disagreement, because marketwide signal shocks are correlated with individual dividend shocks. This is an important feature for understanding the properties of asset returns in our model, according to the recent literature studying the distinct properties of short-run and long-run asset returns.

In a context similar to ours, Dumas, Kurshev, and Uppal (2009) show that the long-run price of risk can be decomposed into two components, a local price of risk and a second term that depends on future disagreement. The intuition for this decomposition is simple. Suppose that at time $t = 0$ disagreement on fundamentals (dividends) is zero, so that the local price of dividend disagreement risk is zero and the instantaneous expected excess return on a risky asset equals that of an equivalent homogeneous economy. Agents know that at a future time $t = T$ they will disagree with probability one. Thus, if disagreement is a state variable in equilibrium, they know that prices at time T will depend on future disagreement. This implies that their expected returns between time 0 and time T depend on their beliefs today about their future dividend disagreement at time T . It follows that higher-order beliefs about future disagreement can affect expected asset returns calculated over discrete time horizons, as opposed to the local case for $dt \rightarrow 0$.

According to our model, higher order dividend disagreement is related to variable $\Psi_z(t)$, for which no simple direct empirical proxy is available, making it challenging to test the relation between asset returns and higher order disagreement. To investigate whether our bottom-up empirical proxy for common disagreement can proxy for higher-order disagreement components, we examine whether it systematically explains future dividend disagreement across firms.¹² To this end, we run the following set of predictive regressions:¹³

$$\Psi_i(t+1) = \alpha + \beta_z \Psi_z(t) + \beta_i \Psi_i(t) + \epsilon(t),$$

where $\Psi_i(t+1)$ ($\Psi_i(t)$) is firm-specific disagreement for firm i at time $t+1$ (t) and $\Psi_z(t)$ is our bottom-up empirical proxy for common market disagreement at time t . Figure 6 plots the estimated slope coefficient $\hat{\beta}_z$ (left panel) across industries, together with the associated t -

statistics. Interestingly, we find that on average our common disagreement proxy significantly predicts future firm disagreement. In Table III, we report the estimated slope coefficients $\hat{\beta}_z$ for different industries, where industries are defined according to GICS codes. We find that estimated coefficients are in most cases positive, so that larger common disagreement today implies higher firm-specific future disagreement.

Figure 6 and Table III about here

For most industries, the estimated coefficient is also significant. When adding lagged firm-specific disagreement in the regressions, the significance of the common disagreement proxy drops, as firm-specific disagreement is a persistent process. However, even in this case, aggregate disagreement is significant for more than 50% of all industries.

These results reveal that our bottom-up proxy for common disagreement is indeed a powerful predictor of future firm-specific disagreement, consistent with the intuition for $\Psi_z(t)$ in our theoretical model.

B. Panel Regression Results

B1. Disagreement and Variance Risk Premia

Model Prediction II states that the absolute difference between index and individual volatility risk premia increases with the degree of heterogeneity in beliefs. Let $IV_i(t) - RV_i(t+1) \equiv VRP_i(t)$ be the difference between the implied and realized volatility of firm i at time t and $t+1$, respectively.¹⁴ Implied volatility $IV_i(t)$ is computed for one-month maturity at-the-money options. Realized volatility $RV_i(t+1)$ is computed as the square root of the sum of squared daily underlying returns over the month. The risk premium, $IV(t) - RV(t+1) \equiv VRP(t)$, denotes the same difference for the index.

According to Model Prediction II, a regression of $VRP(t) - VRP_i(t)$ on the disagreement proxies should yield a positive regression parameter, when controlling for additional risk factors potentially related to volatility risk premia. To test this hypothesis, we estimate the following panel regression model:

$$\begin{aligned} VRP(t) - VRP_i(t) = & \beta_0 + \beta_1 DIB_i(t) + \beta_2 \overline{DIB}(t) + \sum_j \gamma_j \text{Control}_{ij}(t) \\ & + \sum_k \delta_k \text{Control}_k(t) + \epsilon_i(t), \end{aligned} \tag{6}$$

where $DIB_i(t)$ is the proxy for belief disagreement for firm i at time t , $\overline{DIB(t)}$ is the common disagreement proxy at time t , and $\text{Control}_i(t)$ is a vector of control variables for firm i at time t , which contains measures of individual stock liquidity, stock market beta, option buying pressure, and option implied skewness. $\text{Control}(t)$ is a vector of time-series determinants at time t , including market volatility, the macro factor, and the sentiment index. Table IV (first three columns) summarizes the estimation results.

Table IV about here

Consistent with Model Prediction II, parameter estimates in column 1 of Table IV for individual and common disagreement proxies are positive and significant, both economically and statistically. For instance, a one-standard-deviation change in the individual DiB proxy implies almost a three-quarter standard-deviation change in the difference between the index and individual volatility risk premia. To control for potential biases due to spurious time-series correlation, we also add time dummies to the baseline regression (column 2 in Table IV) and find that they have a negligible effect on the estimated coefficients for the individual disagreement proxy. The estimated coefficient for the common disagreement proxy drops by approximately 20% and remains significant at the 10% level. This is consistent with the fact that the common disagreement proxy picks up time-variation effects only. Similar findings apply to the other time-series variables, like market volatility or the macro factor. To learn more about the degree of cross-sectional variation explained, we run a fixed effects panel regression (column 3 in Table IV) and find that the coefficient estimate for the individual DiB proxy is virtually unchanged and still significant. This finding indicates that individual DiB captures time-varying cross-sectional heterogeneity in the difference between index and individual volatility risk premia, rather than a pure time-independent cross-sectional effect.

B2. Disagreement and the Skew of Index and Individual Options

Model Prediction I states that a larger degree of disagreement is linked, *ceteris paribus*, to a more negative difference between the slopes of index and individual option smiles. To test this prediction, we compute a standardized measure of the slope of the smile, both for index and individual options, as the difference between the implied volatility of a put with 0.92 strike-to-spot ratio (or the closest available) and the implied volatility of an at-the-money put, divided by the absolute difference in strike-to-spot ratios. As in the previous section, we

consider one-month maturity options. Model Prediction I implies a positive relation between disagreement proxies and the difference between index and individual slopes of the smile based on this metric. To test this hypothesis, we estimate the following panel regression:

$$\underbrace{Slope(t)}_{\text{Slope Index}} - \underbrace{Slope_i(t)}_{\text{Slope Individual}} = \beta_0 + \beta_1 DIB_i(t) + \beta_2 \overline{DIB(t)} + \sum_j \gamma_j \text{Control}_{ij}(t) + \sum_k \delta_k \text{Control}_k(t) + \epsilon_i(t), \quad (7)$$

where $Slope(t) - Slope_i(t)$ is the difference between index and individual slopes of the smile. We consider the same set of explanatory variables as in panel regression (6), extended by the index and individual stock at-the-money implied volatilities, which are useful to control for the direct impact of the volatility level on the skew of index and individual smiles. Table IV (columns 4 to 6) summarizes the estimation results.

Consistent with Model Prediction I, we find a positive relation between disagreement and the difference between index and individual option slope proxies (column 4 of Table IV). Individual and common DiB proxies are both economically and statistically significant. For instance, a one-standard-deviation change in the individual DiB proxies implies about half a standard-deviation change in the difference between index and individual option slopes. Regressions including time and fixed effects (column 5 and 6 of Table IV) show that the coefficient estimate for individual DiB is not driven by spurious time-series correlation or by simple time-independent cross-sectional features.

B3. Disagreement and the Correlation Risk Premium

Model Prediction II also states that the negative correlation risk premium embedded in the cross-section of index and individual options is positively linked to the common component in the heterogeneity of beliefs. We follow the procedure detailed in Section II.C., to compute observable proxies for the monthly time series of payoffs of a short correlation swap, denoted by $IC(t) - RC(t + 1)$.¹⁵ We compute the realized correlation $RC(t + 1)$ at time $t + 1$ as the average realized correlation of the daily returns of stocks in the S&P 100 index over the previous month. We infer the correlation swap rate $IC(t)$ at time t from the cross-section of one-month maturity index and individual stock variance swap rates, which are synthesized from index and individual stock implied volatility functions at time t .

Model Prediction II states a positive relation between common disagreement and the expected payoff of a short correlation swap. To test this hypothesis, we estimate the following linear regression model:

$$IC(t) - RC(t + 1) = \beta_0 + \beta_1 \overline{DIB}(t) + \sum_k \gamma_k \text{Control}_k(t) + \epsilon(t) , \quad (8)$$

where vector $\text{Control}(t)$ includes as control variables the macro factor and our proxies for market volatility, index option demand pressure, and sentiment. Estimation results are summarized in Table V.

Table V about here

Consistent with Model Prediction II, the common disagreement component has a significant positive impact on the negative correlation risk premium (column 1 of Table V) and is the most significant variable, also after controlling for demand pressure (column 2 of Table V) and sentiment (column 3 of Table V).

C. Option Strategy Returns

Model Prediction III states that (priced) disagreement risk is systematically linked to the returns of sorted straddle and dispersion option portfolios. This link should occur both in the time series and in the cross-section. Therefore, we take the previous results one step further and study the empirical risk-return profile of excess returns generated by these option strategies.

C1. Disagreement and the Time Series of Straddle Portfolio Returns

Goyal and Saretto (2009) consider option strategies that take advantage of cross-sectional differences in stocks' realized and implied volatilities. They show that a straddle portfolio that is long (short) in the portfolio with a large positive (negative) difference between realized and implied volatility yields economically large returns, with no apparent exposure to other standard risk factors. Model Prediction III suggests a novel interpretation of their findings: if the common factor driving stock disagreement is linked to states of active risk sharing among investors, a fraction of these straddle returns might reflect compensation for bearing systematic disagreement risk.

We test empirically this hypothesis and run a similar exercise as in Goyal and Saretto (2009) by constructing quintile-sorted straddle portfolios with respect to the level of individual disagreement. Each month, quintile 1 (5) consists of firms with the lowest (highest) disagreement proxy. For each stock in each quintile, we compute individual straddle portfolio returns. Next, we average the straddle returns within each quintile and obtain equally weighted straddle portfolio returns sorted with respect to individual stock disagreement. We then regress the straddle excess returns on the innovation in common disagreement proxy, $\epsilon^D(t)$, while controlling for other standard risk factors:

$$\begin{aligned} exret_i(t) = & \alpha_i + \beta_i^M MRKT(t) + \beta_i^S SMB(t) + \beta_i^{BM} HML(t) + \beta_i^{MOM} MOM(t) \\ & + \beta_i^V \epsilon^V(t) + \beta_i^L \epsilon^L(t) + \beta_i^S \epsilon^S(t) + \beta_i^C \epsilon^C(t) + \beta_i^D \epsilon^D(t) + u_i(t), \end{aligned} \quad (9)$$

where $exret_i(t)$ is the monthly excess return of quintile straddle portfolio i , $i = 1, \dots, 5$, $MRKT(t)$ is the market excess return and $SMB(t)$, $HML(t)$, and $MOM(t)$ are the size, book-to-market and momentum factors, and $\epsilon^j(t)$, $j = V, L, S, C, D$, is the monthly change in the VIX, liquidity, sentiment, realized correlation, and common disagreement proxies, respectively. Table IV reports the results for straddles in quintiles 1 (low) and 5 (high), together with those of a spread portfolio (LmH), consisting of a long position in quintile 1 straddles and a short position in quintile 5 straddles.

Table VI about here

Consistent with Model Prediction III, we find that the straddle average excess return decreases from 7.2% for quintile 1 to -8.7% for quintile 5, implying an annualized return of 15.9% and a Sharpe ratio of about 1.32 for the spread straddle portfolio. The most significant explanatory variable for spread straddle returns is the innovation in common disagreement, $\epsilon^D(t)$, which has the expected positive coefficient and a t -statistic of 1.98. Standard risk factors are largely insignificant at usual confidence levels (see the third column of Table VI). These include the monthly change in the VIX index $\epsilon^V(t)$ (realized stock correlation $\epsilon^C(t)$), which captures potential time-varying volatility (correlation) risk effects, and the monthly change in the liquidity and sentiment proxies, denoted by $\epsilon^L(t)$ and $\epsilon^S(t)$, which capture the effects of unexpected shocks in overall liquidity and sentiment conditions. Among these variables, only the estimated slope coefficient for the realized correlation is weakly significant at the 10% significance level. Consistent with Model Prediction III, we cannot reject the hypothesis

that straddle portfolio returns contain a systematic return component positively related to unexpected changes in systematic disagreement risk.

C2. Disagreement and the Time Series of Dispersion Portfolio Returns

Dispersion strategies generate a risk exposure to unexpected changes in correlation. Model Prediction III implies a close link between dispersion returns and their exposure to common disagreement shocks. To test this prediction, we start from the simple practical consideration that trading a whole book of constituent volatilities can generate unattractive transaction costs or be difficult due to illiquidity issues; trading a sub-basket of individual stock volatilities by means of so-called proxy hedging, is more appropriate.

We sort individual stock options into five quintiles based on the degree of individual firm disagreement. We then select the stocks in the bottom quintile to build the long side of the dispersion portfolios in our empirical tests. For comparison purposes, we also present results for a dispersion portfolio where the long side consists of options of stocks in the top quintile.¹⁶ Table VI (columns 4 and 5) summarizes the properties of these dispersion returns.

We find that the DiB sorted low quintile dispersion portfolio has the most attractive average return (16.54%) and unconditional Sharpe ratio (1.51). Estimated regression parameters of factor model (9) for dispersion returns reveal additional interesting features. First, dispersion portfolios have no apparent exposure to standard risk factors, such as the market return or the size, book-to-market, and momentum factors, or to liquidity shocks. Second, they feature a slightly significantly negative (positive) exposure to aggregate implied volatility (sentiment) shocks. Third, the most significant positive (negative) exposure arises with respect to shocks in the common disagreement components (average realized stock correlation). The economic impact of common disagreement shocks on the average return of a dispersion portfolio is economically large: for instance, a one-standard-deviation shock implies an increase of the average return of DiB-sorted dispersion portfolios of about 3.5%. Consistent with Model Prediction III, we conclude that (i) dispersion excess returns are not easily explained by standard sources

of risk and (ii) they feature a distinct and economically important exposure to common disagreement shocks.

C3. Double-Sorted Portfolios and the Cross-Section of Dispersion Returns

According to the evidence in Table VI, different exposures to common disagreement shocks in the time series might be linked to systematically different excess returns in the cross-section of dispersion portfolios: dispersion returns in the bottom DiB quintile have both a larger exposure to disagreement shocks and a higher excess return than those in the top DiB quintile. Are these cross-sectional differences significant, after controlling for other standard risk factors, such as, liquidity risk?

We can address this question using a simple nonparametric approach. The advantage of this approach over a cross-sectional regression is that it does not impose a linear structure on returns (the disadvantage is that we can control for only one characteristic at a time). In each month, we construct two-way-sorted straddle portfolios for the long side of the dispersion trades. The first sort is based on the degree of exposure to common disagreement risk. The second sort is with respect to the degree of exposure to other risk factors in factor regression (9). To keep the double-sorted portfolios sufficiently well populated, we double-sort options into terciles, rather than quintiles, along each sorting dimension.

Each column of Table VII reports aggregated returns (and related t -statistics) of double-sorted straddle and dispersion portfolios: returns in each row are obtained by aggregating double-sorted returns along the market, size, book-to-market, momentum, liquidity, and volatility sorting dimension. Therefore, the returns represent returns of disagreement straddle and dispersion portfolios controlling for characteristics.

Table VII about here

We find that both for straddle (Panel A) and dispersion (Panel B), differences in portfolio returns between bottom and top terciles of common DiB exposure are significant and economically relevant, irrespective of the choice for the second sorting criterion. The double-sorted spread portfolio average returns are similar to those reported in Table VI, albeit slightly smaller, which is expected as we are sorting into tercile portfolios instead of quintiles. Overall, these results show that an economically relevant fraction of cross sectional variation in straddle and dispersion returns is driven by conditional exposure to systematic disagreement risk that is not subsumed by other standard sources of risk.

D. The Price of Disagreement Risk

We conclude our empirical study by estimating the market price of disagreement risk in the cross-section of option returns. Following Fama and French (1993), we first construct a factor-mimicking portfolio for common disagreement risk, which is long (short) the quintile of stocks having the highest (lowest) beta with respect to common disagreement shocks. We call the excess return of this factor-mimicking portfolio CDiB. By having such a traded disagreement portfolio at hand, we can study and interpret the factor price of disagreement risk in a more convenient way.

We estimate the market price of common disagreement risk using a two-pass Fama and MacBeth (1973) regression approach. In the first stage of the methodology, we regress the time series of monthly excess returns of straddle and dispersion quintile portfolios on a number of risk factors, in order to estimate their factor betas. In addition to the excess returns of replicating portfolios for standard risk factors, including the market (MRKT), size (SMB), and book-to-market (HML) portfolios, we consider the excess return of our factor-mimicking portfolio for disagreement risk.

In the second stage of the methodology, for each month we regress the cross-section of excess returns of straddle and dispersion portfolios on their estimated factor betas from the first step. Hence, for each month $t = 1 \dots, T$, we estimate the following cross-sectional regressions:

$$exret_i(t) = \alpha_i + \beta_i^{MRKT} \lambda_{MRKT} + \beta_i^{SML} \lambda_{SML} + \beta_i^{HML} \lambda_{HML} + \beta_i^{CDiB} \lambda_{CDiB} + u_i(t), \quad (10)$$

where $exret_i(t)$ is the excess return of an option straddle or dispersion portfolio, β_i^K is the beta with respect to the K th factor, and λ_K is the price of risk for the K th factor-mimicking portfolio. In this way, we obtain monthly estimates of the price of risk for each risk factor. Finally, we average monthly point estimates of prices of risk over time to obtain unconditional estimates of factor risk premia. Table VIII summarizes our findings.

Table VIII about here

Panel A reports estimated prices of risk.¹⁷ Overall, we find that CDiB carries a highly significantly positive price of risk, while the standard risk factors do not. The estimated monthly price of risk is 6.4% for straddle returns and 3.4% for dispersion returns.

Panel B1 (B2) reports the cross-section of estimated betas in the first step of our two-step regression procedure. We find that the CDiB betas of straddle (dispersion) quintile portfolios range from 1.124 to -0.347 (0.624 to 0.005). Thus, cross-sectional variation in factor betas for common DiB are responsible for differences in average excess returns of sorted straddle (dispersion) portfolios of about $(1.124 - (-0.347)) \cdot 0.06 = 9.4\%$ $((0.624 - 0.005) \cdot 0.03 = 2.10\%)$. This amounts to about two-third (one-third) of the total cross-sectional variation of 15% (6%) in the excess returns of straddle (dispersion) portfolios.

In summary, the cross-section of option portfolio returns contains an economically relevant component generated by different cross-sectional exposures to systematic disagreement risk. When neglecting exposures to common disagreement components, average alphas of dispersion and straddle returns are large and significantly different from zero.

V. Conclusions

We motivate theoretically and test empirically the equilibrium relation between investors' disagreement, the market price of volatility and correlation, and the differential pricing of index and individual equity options.

In a two-tree Lucas economy with unknown firm growth rates and heterogeneous perceptions of aggregate uncertainty across investors, we propose a new rational explanation for the differential pricing of index and individual equity options. Agents agree to disagree and engage in risk-sharing to finance their optimal consumption plans. In this setting, investor disagreement about aggregate economic news can generate substantial stock return volatility and comovement, even when underlying fundamentals are weakly related. The effect is state dependent and follows from the increased risk-sharing between agents when subjective uncertainty and disagreement are larger. More active risk-sharing against bad aggregate economic states generates an endogenous correlation risk premium that implies both a greater index volatility risk premium than for individual stocks and a steeper option smile of index relative to individual stock options. Option volatility and correlation portfolios also imply large risk-adjusted excess returns, due to their implicit exposure to disagreement risk, with no apparent exposure to other standard risk factors.

Using analysts forecast data on future earnings to construct proxies for belief disagreement, we find that disagreement has strong explanatory power for (i) the spread between index and

individual volatility risk premia, (ii) the difference between the slope of index and individual option smiles, and (iii) option-implied correlation risk premia. We also find that option volatility and correlation portfolios yield returns with attractive Sharpe ratios, no significant exposure to standard risk factors, but an economically and statistically significant exposure to systematic disagreement shocks.

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Notes

¹A dispersion portfolio involves a short position in the index volatility and a long position in the constituents volatility, which can be based on either straddles or strangles. Since at-the-money straddles have a delta exposure close to zero, a dispersion portfolio short an index straddle and long the constituents straddles is well hedged against market movements.

²A formal proof of this result can be found in Liptser and Shiryaev (2000).

³For simplicity of interpretation, we consider a symmetric economy with identically distributed dividends across firms.

⁴See Cox and Huang (1989). The extension to the case with heterogeneous beliefs is due, among others, to Cuoco and He (1994), Karatzas and Shreve (1998), and Basak and Cuoco (1998). In this extension, the utility function of the representative agent is a weighted average of the utility functions of the individual agents. In contrast to the standard setting, a stochastic relative weight $\lambda(t)$ captures the equilibrium effect of the heterogeneity in beliefs across agents.

⁵For convenience, we compute all pricing expressions from the perspective of agent A , using the first good as a numéraire.

⁶We set $\Psi_{D_2}(t) = 0$ for simplicity, which is a conservative assumption for the implied correlation level.

⁷If one can only observe stock correlations (as opposed to the correlation of the differences in beliefs across stocks), the average stock return correlation will emerge as a risk factor explaining (in reduced-form regressions) the spread between the volatility risk premium of index and individual options. A related reduced-form argument is used by Driessen, Maenhout, and Vilkov (2009) to explain the difference between index and individual volatility risk premia by the presence of a large (negative) correlation risk premium.

⁸Instead of purchasing the whole basket of constituent straddles, a dispersion strategy can be enhanced by selecting a subset of individual options with low implied volatility relative to the expected volatility. Since expected volatility is unobservable, correlation option traders often proxy for it using the current realized volatility, because volatilities feature a substantial degree of autocorrelation.

⁹For instance, if a forecast is made in July and last confirmed in September we use this information for the months July, August, and September. To exclude stale forecasts, we delete forecasts that are outstanding more than 90 days from issuance. A common disagreement proxy computed as the equally weighted average of individual firm disagreement has a correlation of almost one with the proxy based on market capitalization weights.

¹⁰These data are available on Jeffrey Wurgler's webpage, <http://pages.stern.nyu.edu/~wurgler>.

¹¹Since the price-to-earning ratio for the S&P 100 does not exist, we use price-to-earning data from the S&P 500. We retrieve S&P 500 price-to-earnings data from the S&P webpage, while all other macro variables are from FRED.

¹²We thank an anonymous referee for suggesting this interpretation of our model.

¹³All variables are standardized to have a mean of zero and a risk-neutral deviation of one.

¹⁴For brevity, we denote $IV_i(t, t+1)$ by $IV_i(t)$ and $RV_i(t, t+1)$ by $RV_i(t+1)$.

¹⁵For brevity, we denote $IC(t, t+1)$ by $IC(t)$ and $RC(t, t+1)$ by $RC(t+1)$.

¹⁶We fix the portfolio weights of the individual options by hedging away individual volatility risks using a vega-neutral position. We then determine delta-neutral portfolio weights for the individual stocks and invest the residual wealth in the risk-free asset. Thus, each dispersion portfolio shorts an (index) straddle and invests the residual wealth in (individual) straddles, individual stocks, and the risk-free bond. We circumvent microstructure biases and initiate all option strategies on Tuesdays, as opposed to the first trading day (Monday). Returns are computed using as a reference initial price the average of the closing bid and ask quotes on the previous day. As a reference final price for all options in the portfolio, we compute the realized value of the terminal payoff at the option's expiration date. After expiration of each option, a new option is selected and a new monthly return is calculated following the above procedure. We use equally weighted monthly returns on calls and puts. Such a procedure is repeated for each month.

¹⁷We adopt standard refinements of the two-pass methodology, using the Shanken (1992) correction to obtain errors-in-variables robust standard errors. We also compute the Shanken and Zhou (2007) correction, to obtain standard errors robust against misspecification. Our results are unchanged.

Table I
Parameter Values and Benchmark Values for State Variables

This table lists the parameter values used for all figures in the paper. We calibrate the model to the mean and volatility of the dividends on the S&P 500. The average growth rate for the 1996 to 2006 period is 5.93% and the volatility is 3.52%. The initial values for the conditional variances are set to their steady-state variances. We study a symmetric economy, where parameters for firms 1 and 2 are assumed to be the same.

Parameters for Fundamentals		
Long-term growth rate of dividend growth	a_{0D_i}	0.01
Mean-reversion parameter of dividend growth	a_{1D_i}	-0.01
Volatility of dividend	σ_{D_i}	0.07
Initial level of dividend	D_i	1.00
Initial level of dividend growth	$m_{D_i}^A$	0.01
Parameters for Signal		
Long-term growth rate of signal	a_{0z}	0.01
Mean-reversion parameter of signal	a_{1z}	-0.03
Volatility of signal	σ_z	0.06
Agent specific Parameters		
Relative risk aversion for both agents	γ	2.00
Time Preference Parameter	δ	0.02
Difference in economic uncertainty	$\sigma_{\mu_z}^A - \sigma_{\mu_z}^B$	-0.01
Average economic uncertainty	$0.5 (\sigma_{\mu_z}^A + \sigma_{\mu_z}^B)$	0.02

Table II
Moments of Simulated Option Strategies

This table reports summary statistics of the simulated strategy returns: mean, standard deviation, skewness, kurtosis, and Sharpe ratio. The DiB-sorted straddle portfolio is long the option with the lower disagreement and short the option with the higher disagreement. The DiB-sorted dispersion portfolio is formed by investing 100% of the wealth in shorting index straddles, investing a fraction of wealth into the options of the firm with either the lowest or the highest belief disagreement such that the portfolio is vega neutral, and invest the remainder in the individual stock of this particular firm in a way that makes the portfolio delta neutral. The index consists of two equally weighted stocks. We simulate 2,877 trading days and 1,000 simulation runs. All options are at-the-money with a maturity of 28 trading days. We also run the regressions: $exret_i(t) = \alpha_i + \beta_i^M MRKT(t) + \beta_i^D \epsilon^D(t) + \beta_i^{(\epsilon^D)^2} (\epsilon^D(t))^2 + u_i(t)$, where $exret_i(t)$ is the return in excess of the one month risk-free rate, $MRKT(t)$ is the index excess return, $\epsilon^D(t)$ is the monthly change in the common disagreement factor, and $(\epsilon^D(t))^2$ is the monthly change in the common disagreement squared. The market return is defined as the return on the index (i.e., the equal weighted sum of both stocks) and the common disagreement is the weighted average of the individual dividend disagreement proxies, Ψ_{D_i} . We set the inattention parameters α_{D_i} equal to 0.1.

	DiB Strdl	Low DiB Disp	High DiB Disp	Short Put	Index
Mean	0.081	0.127	0.087	0.117	0.010
StDev	0.273	0.231	0.242	0.653	0.038
Kurtosis	6.938	7.532	7.146	8.477	5.819
Skewness	1.102	-3.127	-4.154	-8.239	-7.423
Sharpe ratio	1.03	1.90	1.25	0.62	0.91
Alpha	0.081 (1.77)	0.079 (1.89)	0.065 (1.87)	0.091 (2.02)	
MRKT	1.942 (1.25)	-0.975 (-0.99)	-0.889 (-1.04)	4.290 (2.00)	
ϵ^D	2.784 (2.67)	4.105 (3.04)	3.475 (2.48)	3.144 (2.07)	
$(\epsilon^D)^2$	-0.081 (-1.88)	-0.104 (-1.61)	-0.214 (-1.55)	-0.115 (-1.64)	

Table III
Common Disagreement Predictability Regressions

This table reports estimated coefficients from regressing firm-specific disagreement on common disagreement:

$$\Psi_i(t+1) = \beta_z \Psi_z(t) + \beta_i \Psi_i(t) + \epsilon(t),$$

where $\Psi_i(t+1)$ is firm-specific disagreement for firm i at time $t+1$, $\Psi_z(t)$ is common disagreement at time t , and $\Psi_i(t)$ is firm-specific disagreement at time t . We average firm disagreement according to industry type, where industry is defined according to GICS codes. t -statistics presented in parentheses are calculated following Newey and West (1987). All variables are standardized to have mean zero and a standard deviation of one. Data are monthly and run from 1996 through 2007.

Industry	1010	1510	2020	2030	2510	2520	2530	2540	2550	3020
Common DiB	0.268 (2.65)	0.387 (3.53)	0.083 (0.93)	0.331 (3.29)	0.310 (3.03)	0.338 (2.62)	0.447 (3.65)	0.184 (1.87)	0.322 (2.93)	0.033 (0.34)
Adj. R^2	6.95%	14.49%	0.67%	10.57%	9.36%	11.06%	19.31%	3.29%	10.00%	0.11%
lagged	0.998 (127.80)	0.888 (21.68)	0.754 (10.41)	0.743 (30.07)	0.793 (15.11)	0.970 (53.90)	0.895 (13.63)	0.594 (6.04)	0.946 (36.01)	0.657 (15.29)
Common DiB	0.000 (0.02)	0.056 (1.64)	-0.003 (-0.07)	0.071 (1.58)	0.075 (3.66)	0.007 (0.26)	0.129 (4.66)	0.124 (2.78)	0.043 (2.30)	0.007 (0.13)
Adj. R^2	96.78%	82.44%	56.59%	58.62%	67.18%	88.25%	83.49%	37.12%	87.15%	42.35%
Industry	3030	3510	3520	4010	4020	4030	4510	4520	5010	5510
Common DiB	0.373 (4.53)	0.032 (0.31)	0.134 (1.15)	0.330 (1.94)	0.271 (1.85)	0.174 (1.46)	0.369 (3.23)	0.368 (3.03)	0.297 (3.34)	0.246 (2.49)
Adj. R^2	13.45%	0.10%	1.76%	10.52%	7.15%	2.97%	13.25%	13.13%	8.47%	5.83%
lagged	0.852 (20.91)	0.909 (29.37)	0.943 (43.30)	1.174 (12.27)	1.072 (11.57)	0.993 (49.62)	0.966 (38.19)	0.928 (56.17)	0.712 (8.83)	0.923 (36.94)
Common DiB	0.088 (5.90)	0.001 (0.03)	0.030 (0.66)	0.050 (1.55)	0.095 (2.03)	0.026 (1.12)	0.053 (1.66)	0.086 (3.74)	0.096 (2.24)	0.045 (2.83)
Adj. R^2	78.25%	82.73%	87.38%	90.49%	86.15%	94.29%	90.68%	90.71%	54.81%	86.51%

Table IV
Difference between Index and Individual Volatility Risk Premium and Slope

Using data from January 1996 to June 2007, we run panel regressions of proxies for the difference between index and individual volatility risk premia (and the slope) on a number of determinants. DiB is our proxy for difference in beliefs for each firm, computed from the mean absolute difference among analysts' forecasts. Common DiB is our proxy for the common component in the difference in beliefs, Market vola is the 21-day realized volatility of the index, and Skewness is the difference between the implied volatility of a put with 0.92 strike-to-spot ratio (or the closest available) and the implied volatility of an at-the-money put, dividend by the absolute difference in strike-to-spot ratios. CAPM beta is estimated from a regression using a window of 180 daily returns. Liquidity is the ratio between trading volume and shares outstanding. DP is demand pressure, defined as the difference between the number of contracts traded during the month at prices higher than the prevailing bid-ask quote midpoint and the number of contracts traded during the month at prices below the prevailing bid-ask quote midpoint, times the absolute value of the option's delta, then scaled by the total trading volume across all option series. Macro factor is a dynamic factor from industrial production (IP), housing starts, S&P 500, price-earnings ratio (P/E), and the producer price index (PPI). Sentiment is the first principal component from trading volume as measured by NYSE turnover, the dividend premium, the closed-end fund discount, first-day returns on IPOs, and the equity share in new issues. *, **, *** denotes significance at the 10% , 5%, and 1% level, respectively. All estimations use autocorrelation and heteroskedasticity-consistent t -statistics reported in parentheses below the estimated coefficient.

	Difference in VRP			Difference in Slope		
Constant	0.009*** (4.10)	0.008*** (3.85)	0.009*** (4.01)	0.003*** (15.68)	0.004*** (16.14)	0.004*** (15.87)
DiB	0.131*** (3.94)	0.127*** (4.28)	0.127*** (4.14)	0.104*** (4.51)	0.105*** (5.00)	0.106*** (4.89)
Common DiB	0.099** (2.50)	0.082** (2.22)	0.075** (2.41)	0.101** (2.48)	0.072** (2.06)	0.077** (1.99)
Market vola	0.104** (2.12)	0.092** (1.97)	0.092** (1.98)	0.422** (2.51)	0.399** (2.43)	0.386** (1.99)
ATM vol index				0.422*** (6.29)	0.402*** (5.18)	0.399*** (5.10)
ATM vol individual				-0.107*** (-3.02)	-0.1*** (-2.94)	-0.097*** (-2.98)
Macro factor	-0.098** (-1.99)	-0.101* (-1.73)	-0.101* (-1.76)	-0.105* (-1.84)	-0.089* (-1.74)	-0.086* (-1.73)
Liquidity	-0.001 (-0.91)	-0.001 (-0.61)	-0.001 (-0.60)	-0.001 (-0.01)	-0.001 (-0.02)	-0.001 (-0.03)
CAPM beta	0.021* (1.83)	0.018* (1.80)	0.017* (1.81)	0.010* (1.76)	0.009* (1.67)	0.008* (1.65)
Skewness	-0.001 (-1.03)	-0.001 (-0.83)	-0.001 (-0.82)	-0.001 (-1.03)	-0.001 (-0.83)	-0.001 (-0.82)
DP calls	-0.043 (-1.61)	-0.032 (-1.20)	-0.033 (-1.19)	-0.002 (-1.07)	-0.004 (-1.04)	-0.007 (-1.03)
DP puts	-0.082 (-1.19)	-0.010 (-0.38)	-0.009 (-0.39)	-0.007 (-0.89)	-0.007 (-0.88)	-0.006 (-0.86)
Sentiment	0.101* (1.73)	0.097* (1.80)	0.099* (1.81)	0.102* (2.01)	0.089* (1.76)	0.090* (1.75)
Adj. R^2	0.25	0.20	0.20	0.33	0.32	0.31
Year Dummies		x	x		x	x
Firm Dummies			x			x
Observations	11,318	11,318	11,318	11,318	11,318	11,318

Table V
Correlation Risk Premium Regressions

Using data from January 1996 to June 2007, we run regressions from the correlation risk premium on a number of determinants (see equation (13)). The correlation risk premium is measured using synthetic correlation swaps. Details on the construction are found in the Internet Appendix. Common DiB is our proxy for the common component in the difference in beliefs, Market vola is the 21-day realized volatility of the index, and Macro factor is a dynamic factor from IP, Housing Starts, S&P 500, P/E ratio, and PPI. Demand pressure is the ratio of the open interest for out-of-the-money index put options to the open interest for near- and at-the-money index options, and Sentiment is the first principal component from trading volume as measured by NYSE turnover, the dividend premium, the closed-end fund discount, first-day returns on IPOs and the equity share in new issues. *, **, *** denotes significance at the 10%, 5%, and 1% level, respectively. All estimations use autocorrelation and heteroskedasticity-consistent t -statistics reported in parenthesis below the estimated coefficient.

	(1)	(2)	(3)
Constant	0.002*** (3.45)	0.002*** (3.47)	0.001*** (3.78)
Common DiB	0.104*** (2.71)	0.112*** (2.65)	0.108*** (2.74)
Market vola	0.057** (2.45)	0.068** (2.21)	0.055** (2.01)
Macro factor	-0.079 (-1.49)	-0.065 (-1.41)	-0.065 (-1.12)
Demand pressure		0.875 (0.89)	
Sentiment			0.888** (2.46)
Adj. R^2	0.10	0.09	0.13
Observations	138	138	138

Table VI
Returns on Straddles and Dispersion Sorted on Disagreement

This table reports the monthly mean and annualized Sharpe ratio of the straddle and dispersion portfolio returns. Portfolios are sorted according to the size of each firm's individual DiB. Quintile 1 (5) consists of stocks with the lowest (largest) DiB. Option returns of single stocks are sampled between January 1996 and June 2007. The alpha and beta coefficients are estimated from the least squares regression

$$\begin{aligned} exret_i(t) = & \alpha_i + \beta_i^M MRKT(t) + \beta_i^S SMB(t) + \beta_i^{BM} HML(t) + \beta_i^{MOM} MOM(t) \\ & + \beta_i^V \epsilon^V(t) + \beta_i^L \epsilon^L(t) + \beta_i^S \epsilon^S(t) + \beta_i^D \epsilon^D(t) + \beta_i^C \epsilon^C(t) + u_i(t), \end{aligned}$$

where $exret_i(t)$ is the strategy return in excess of the one month LIBOR, $MRKT$ is the value-weighted excess return on all NYSE, Amex, and NASDAQ stocks, SMB is the size factor, HML is the book-to-market factor and MOM is the momentum factor. ϵ^V is the monthly change in the VIX, ϵ^L is the monthly change in the aggregate liquidity measure, ϵ^D is the change in the common disagreement factor, and ϵ^C is the change in the realized correlation.

	Straddles			Dispersion		
	Low	High	LmH	Low	High	LmH
Mean	0.072	-0.087	0.159	0.165	0.104	0.062
Sharpe ratio	0.61	-1.56	1.33	1.51	1.19	1.10
Alpha	0.083** (2.01)	-0.104* (-1.75)	0.186* (1.92)	0.094 (2.58)	0.067 (2.04)	0.027 (1.87)
MRKT	0.928 (1.02)	-1.192 (-1.24)	2.120 (1.04)	-0.060 (-0.42)	-0.103 (-0.33)	0.043 (0.51)
SMB	-1.293* (-1.93)	1.823 (1.02)	-3.116 (-0.14)	1.621 (1.20)	1.384 (1.39)	0.237 (1.35)
HML	-0.193 (-0.33)	-0.299 (-0.92)	0.106 (0.94)	-0.639 (-0.50)	-0.552 (-0.99)	-0.087 (-1.01)
MOM	0.102 (0.41)	-0.373* (-1.84)	0.475 (1.38)	-1.301 (-0.39)	-1.208 (-1.08)	-0.093 (-0.87)
ϵ^D	0.934** (2.12)	0.680* (1.67)	0.252** (1.98)	4.841*** (2.58)	4.142** (2.14)	0.699** (1.99)
ϵ^L	1.001 (0.83)	0.098 (0.97)	0.903 (1.02)	-0.202 (-0.22)	-0.372 (-0.28)	0.170 (0.54)
ϵ^V	-0.195 (-1.04)	1.235 (1.03)	-1.430 (-0.55)	-2.184* (-1.84)	-2.842* (-1.73)	0.658* (1.79)
ϵ^S	0.287* (1.78)	0.302* (1.93)	-0.015 (-1.03)	3.833* (1.90)	2.377** (2.01)	1.456** (2.39)
ϵ^C	-0.183* (-1.82)	-0.199* (-1.77)	0.016* (1.69)	-0.005*** (-3.01)	-0.011*** (-2.45)	0.006** (1.97)
Adj. R^2	0.08	0.09	0.08	0.05	0.06	0.06
Observations	132	132	132	132	132	132

Table VII
Returns on Double-Sorted Option Strategies

This table reports in Panel A (Panel B) average monthly returns of double-sorted straddle (dispersion) portfolios. Dispersion portfolios are formed by investing 100% of the wealth in shorting index straddles and a fraction of wealth in double-sorted individual stock option portfolios. Each month, we double-sort stocks into nine portfolios using terciles of exposure to Common DiB and terciles of exposure to the market return, size, book-to-market, momentum, aggregate liquidity, and volatility factor, respectively. The returns on the nine double-sorted straddle and dispersion portfolios are then averaged over each of the six stock characteristic portfolios, to obtain three DiB tercile option portfolios controlling for stock characteristics. *t*-statistics reported in parentheses are based on Newey and West (1987) adjusted standard errors using 12 lags. The returns are sampled between January 1996 and June 2007.

Panel A: Average Returns Straddle				
	Low	Med	High	LmH
MRKT	0.064** (1.98)	0.024* (1.87)	-0.054** (-2.01)	0.118** (2.43)
SMB	0.057 (1.65)	0.041** (2.47)	-0.021** (-2.05)	0.078** (2.44)
HML	0.074** (2.55)	0.002** (2.47)	-0.033*** (-3.01)	0.107** (2.48)
MOM	0.087** (1.69)	0.014** (2.45)	-0.001** (-2.33)	0.088*** (2.71)
ϵ^L	0.082** (2.22)	0.014* (1.89)	-0.011*** (-2.64)	0.093** (2.03)
ϵ^V	0.105*** (3.01)	0.025** (2.65)	-0.029** (-2.49)	0.134*** (3.15)
Panel B: Average Returns Dispersion				
MRKT	0.155** (2.23)	0.135*** (2.92)	0.099*** (3.30)	0.056*** (4.30)
SMB	0.155* (1.80)	0.135** (2.01)	0.111* (1.94)	0.044** (2.17)
HML	0.162*** (3.20)	0.134*** (4.30)	0.114*** (3.40)	0.048*** (4.04)
MOM	0.129* (1.76)	0.117** (2.31)	0.093*** (2.67)	0.036*** (2.70)
ϵ^L	0.161** (2.50)	0.148*** (3.74)	0.124*** (4.11)	0.037*** (3.87)
ϵ^V	0.164*** (4.20)	0.146*** (4.11)	0.145*** (5.30)	0.019*** (5.11)

Table VIII
Price of Disagreement Risk

Panel A reports Fama and MacBeth (1973) factor premia of sorted straddle and dispersion portfolios. *MRKT* is the excess return on the market portfolio, *SMB* and *HML* are the Fama and French (1993) size and value factors, and *CDiB* is the mimicking factor for common disagreement. *CDiB* is constructed as a zero cost portfolio that is long the quintile of stocks with highest exposure to Common DiB and short the quintile stocks with lowest exposure to Common DiB. In Panel B1 (B2), we report estimated betas of the *MKT* and *CDiB* factors from the time-series regressions for straddle (dispersion) portfolios. *t*-statistics are corrected for estimation error in the first stage (Shanken (1992)) and are reported in parentheses. The data cover January 1996 until June 2007.

Panel A: Fama and MacBeth (1973) Factor Premia						
	Alpha	MRKT	SMB	HML	CDiB	Adj. R^2
Straddles	0.008 (1.51)	0.005 (0.88)	-0.012 (-1.25)	-0.007 (-0.33)	0.064*** (2.58)	0.76
Dispersion	0.002* (1.67)	0.003 (1.01)	-0.027 (-1.41)	-0.012 (-0.74)	0.034*** (2.44)	0.61
Panel B1: Factor Exposure Straddles						
	1	2	3	4	5	LmH
β_{MRKT}	0.124 (1.01)	0.094 (1.42)	-0.004 (-0.45)	-0.005 (-1.11)	-0.010 (-0.87)	0.134 (1.03)
β_{CDiB}	1.124*** (3.01)	0.100** (2.12)	-0.007* (-1.84)	-0.104** (-2.07)	-0.347*** (-2.65)	1.471*** (2.94)
Panel B2: Factor Exposure Dispersion						
	1	2	3	4	5	LmH
β_{MRKT}	0.214 (0.12)	0.104 (0.84)	0.008 (0.45)	0.001 (0.87)	-0.014 (-0.97)	0.213 (0.83)
β_{CDiB}	0.624*** (3.45)	0.442** (2.45)	0.247*** (2.74)	0.114 (1.54)	0.005** (2.00)	0.619** (2.49)

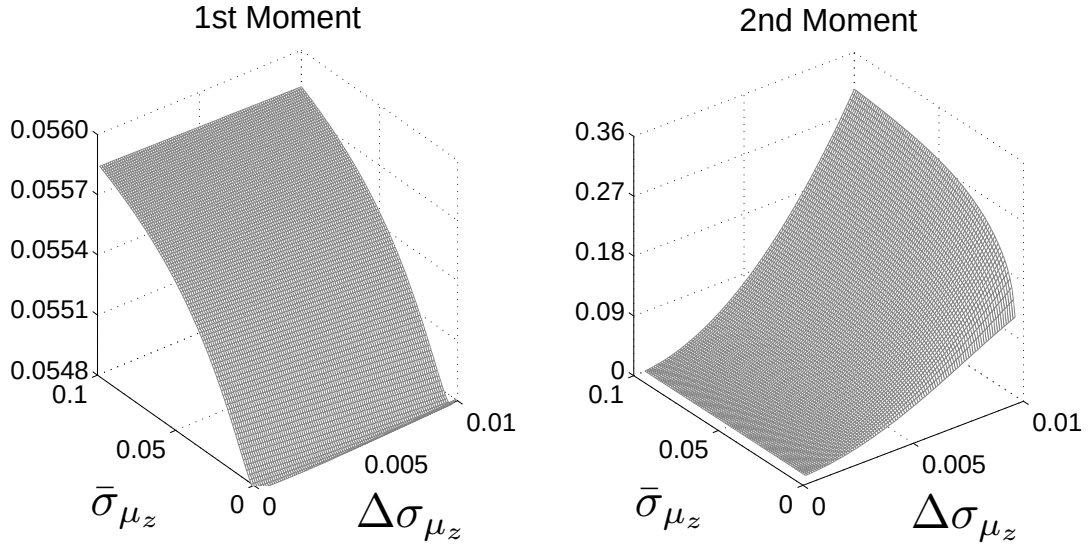


Figure 1. Disagreement as a function of average uncertainty and difference in uncertainty We plot the mean and standard deviation of the distribution of Ψ_{D_1} as a function of the average uncertainty, that is, $\bar{\sigma}_{\mu_z} \equiv \frac{1}{2} (\sigma_{\mu_z}^A + \sigma_{\mu_z}^B)$, and the difference in perceived uncertainty, that is, $\Delta\sigma_{\mu_z} \equiv \sigma_{\mu_z}^B - \sigma_{\mu_z}^A$.

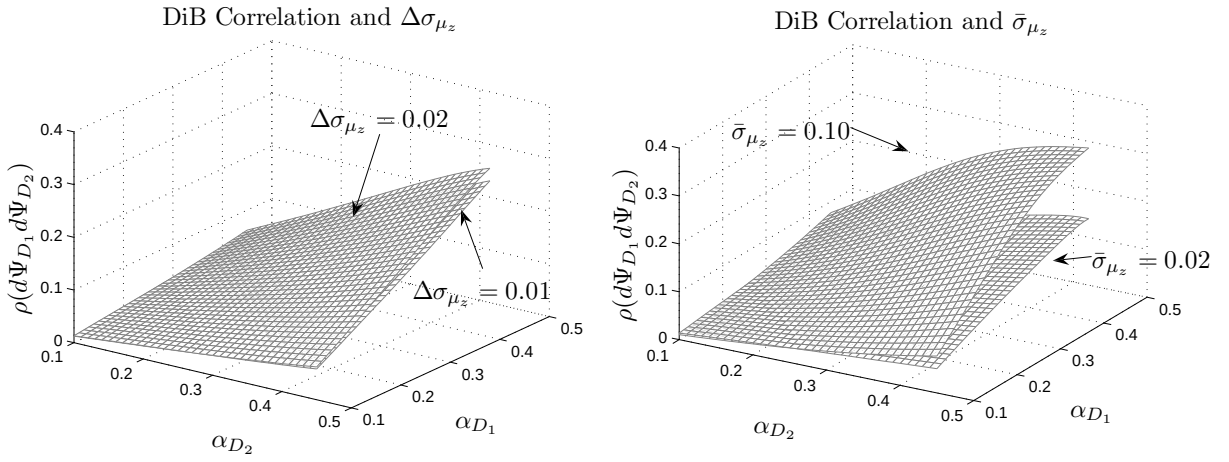


Figure 2. Uncertainty correlation. The left panel plots the instantaneous correlation between the disagreement about firm 1, Ψ_{D_1} , and firm 2, Ψ_{D_2} , as a function of the weights α_{D_1} and α_{D_2} and for different levels of difference in subjective uncertainty ($\sigma_{\mu_z}^B - \sigma_{\mu_z}^A \equiv \Delta\sigma_{\mu_z}$). The right panel plots the instantaneous correlation between the disagreement about firms 1 and 2 as a function of the weights α_{D_1} and α_{D_2} and different levels of average subjective uncertainty ($\frac{1}{2}(\sigma_{\mu_z}^A + \sigma_{\mu_z}^B) \equiv \bar{\sigma}_{\mu_z}$)

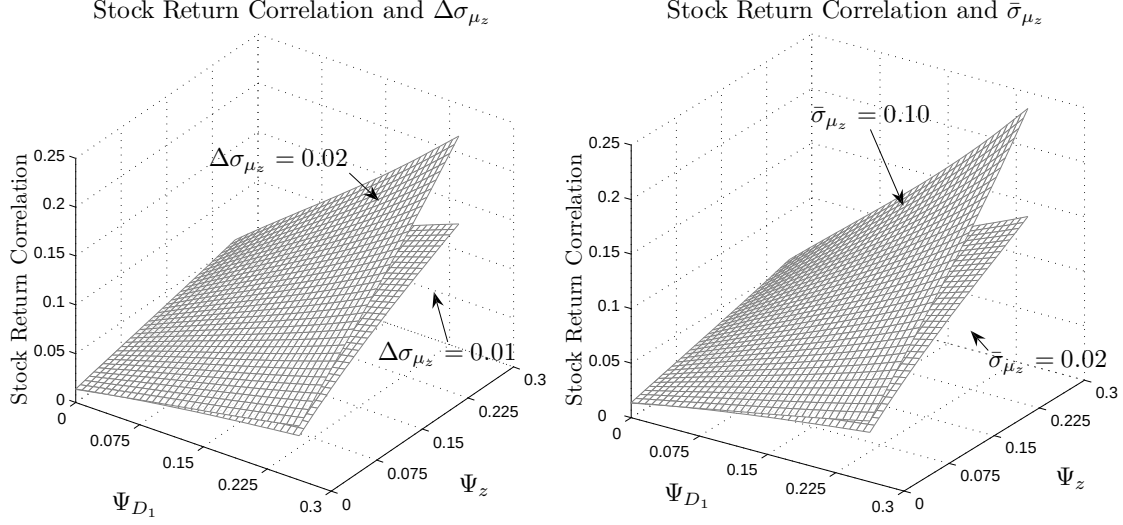


Figure 3. Stock return correlation. This figure plots the return correlation of stocks 1 and 2 as a function of belief disagreement Ψ_{D_1} and Ψ_z for different levels of difference in subjective uncertainty ($\sigma_{\mu_z}^B - \sigma_{\mu_z}^A \equiv \Delta\sigma_{\mu_z}$) (left panel) and different levels of average subjective uncertainty ($\frac{1}{2}(\sigma_{\mu_z}^A + \sigma_{\mu_z}^B) \equiv \bar{\sigma}_{\mu_z}$) (right panel). We set the weights $\alpha_{D_1} = \alpha_{D_2} = 0.1$.

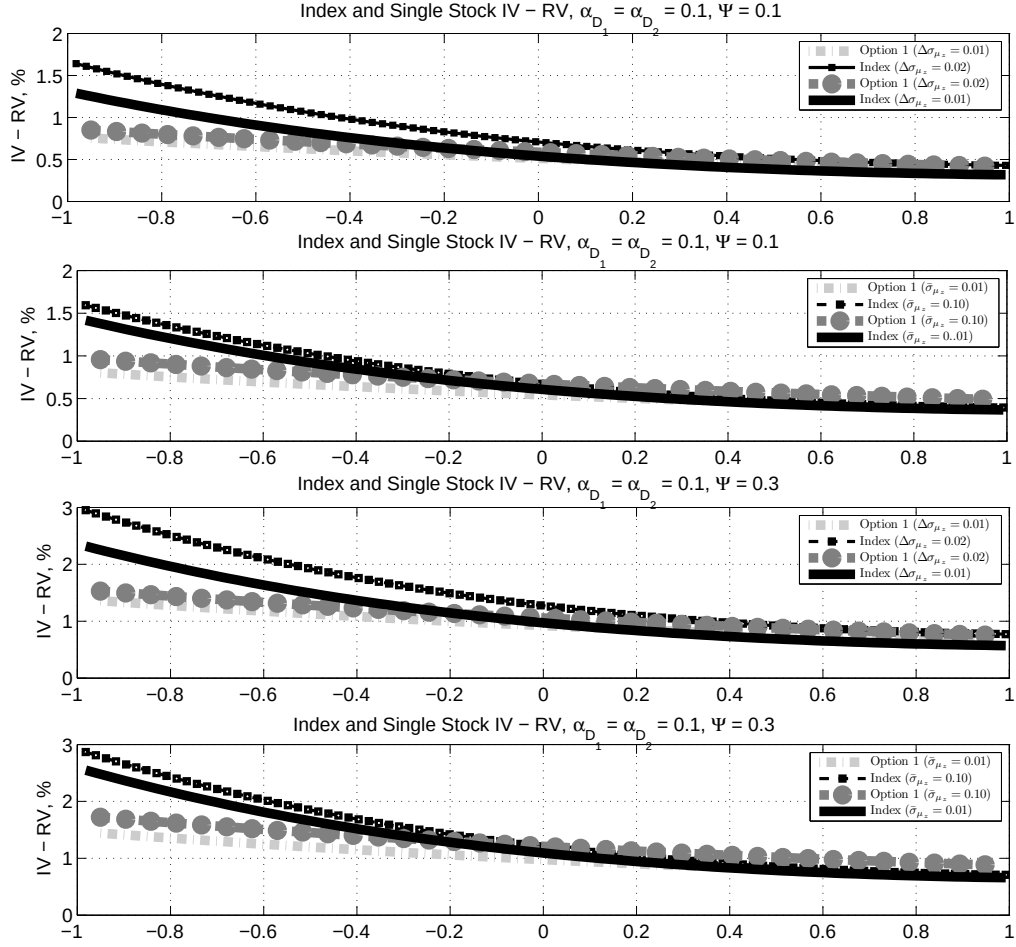


Figure 4. Index and individual IV - RV. This figure plots the volatility risk premium for the individual stock and the index as a function of the moneyness of the option for different levels of disagreement, average uncertainty, and the difference in subjective uncertainty. The volatility risk premium is defined as the difference between the implied volatility and the square root of the expected integrated variance $E_t \left(\int_t^T \sigma^2(s) ds \right)$ of the underlying. Moneyness is defined as $\ln \left(\frac{S}{K} \right)$.

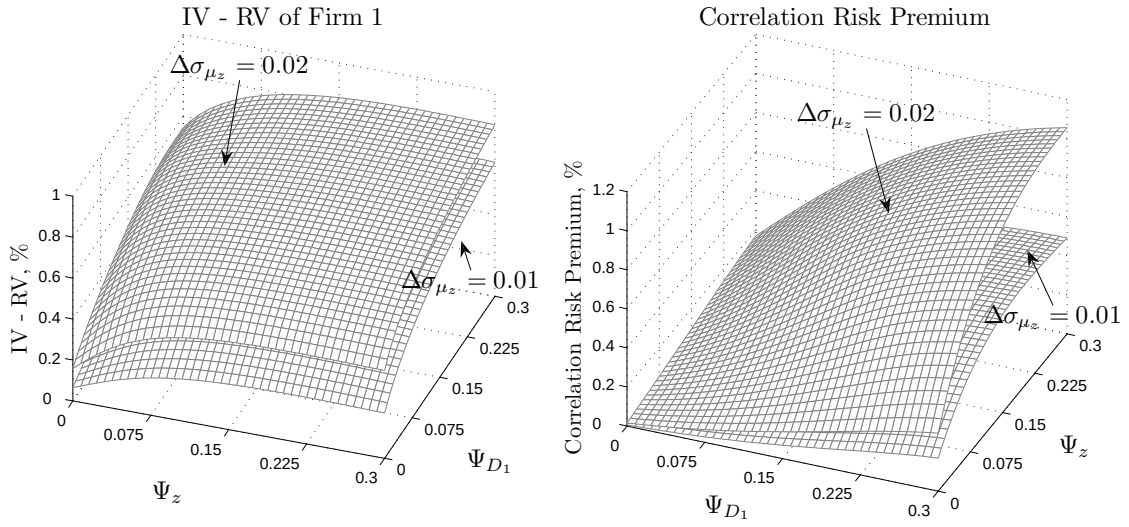


Figure 5. Volatility and correlation risk premia. The left panel plots the volatility risk premium for firm 1 as a function of the disagreement about the growth rate of firm 1, Ψ_{D1} , and the disagreement about the signal, Ψ_z . The volatility risk premium is calculated as the difference between the volatility swap rate and the average 30-day realized volatility. The average 30-day realized volatility is calculated from running 10,000 simulations and averaging. The right panel plots the correlation risk premium, which is synthesized using correlation swaps and the 30-day stock correlation, calculated from running 10,000 simulations.

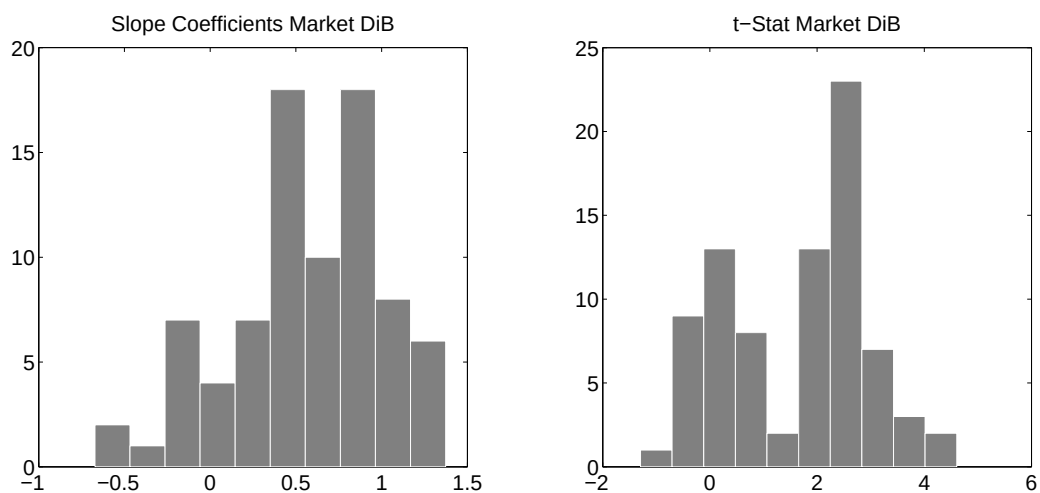


Figure 6. Univariate regression coefficients and t-statistics. This figure plots estimated coefficients and t -statistics from univariate predictive regressions of firm-specific disagreement on common disagreement. All variables are standardized, that is, they have a mean of zero and a standard deviation of one, so that regression intercepts are zero. Data are monthly and run from 1996 through 2007.